

Surrogate-assisted local conditional distribution inference with certified simultaneous gains

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Abstract

Average estimation variance and simultaneous band width can favour different borrowing weights. For selectively verified reference outcomes, we certify gains in simultaneous inference for a local conditional distribution. Paired chooses a common weight between a surrogate-free Anchor and a surrogate-assisted estimator using a lower confidence bound on their Gaussian critical-value difference. The paired gain vanishes at Anchor, and its covariance-estimation error decreases with the borrowing weight. This cancellation yields an Anchor-adaptive oracle bound that allows nonunique optimal weights. Multiplier calibration and independent numerical guards support simultaneous inference after selection and dependence-valid aggregation across sample rotations. A local binary submodel identifies the reference-information scale for certifying weak improvements. Simulations compare Trace, Plug-in, Projection and Paired using common reference samples. An air-monitoring application separates historical mean calibration from subsequent distributional verification and quantifies the reference budget for recovering humidity-specific exceedance discrepancies.

Keywords: Augmented estimation; Conditional distributions; Gaussian approximation; Negative transfer; Selective verification; Simultaneous inference.

1 Introduction

Mean calibration leaves upper-tail agreement unexamined when a cheap instrument accompanies a reference instrument. The air-monitoring study uses a linear calibration trained on historical collocations. In the subsequent audit period, its mean difference is -0.191 micrograms per cubic metre near 30% relative humidity (RH). The calibrated sensor assigns 6.068% of local observations above concentration 15, against 9.307% for the reference instrument. Distributional verification asks which discrepancies can be established from a limited reference sample.

Covariates X and a surrogate S are widely observed, while a reference outcome Y is selectively verified. The target is the conditional cumulative distribution function (CDF) of Y at a chosen x_0 . Augmented inverse-probability weighting remains valid when the fitted outcome regression is inaccurate. A common borrowing path joins surrogate-free Anchor to a surrogate-assisted estimator.

For a simultaneous band, average variance is the wrong criterion. The band's critical value depends on the joint covariance across thresholds, including how the correction redistributes variance. Proposition 1 gives a valid two-threshold path whose population trace-minimising weight reduces total variance and increases the simultaneous critical value. Estimating a band-optimal weight adds covariance uncertainty, since a small apparent reduction may reflect sampling noise.

Surrogate-outcome and two-phase sampling establish the observed-data representation for selectively measured outcomes (Pepe, 1992; Robins et al., 1994; Tao et al., 2017; Tsiatis, 2006). Adaptive augmentation and semi-supervised inference optimise auxiliary-information use for parameter estimation (Cao et al., 2009; Chakraborty and Cai, 2018; Zhang et al., 2019; Song et al., 2024). Prediction-powered inference places this construction in a broad prediction-assisted framework (Angelopoulos et al., 2023a,b; Zrnic and Candès, 2024). Related work documents the finite-reference cost of estimating borrowing weights (Mani et al., 2025; Eyre and Madras, 2025; Kallus and Mao, 2025). These procedures choose auxiliary information for parameter estimation; the selection criterion for a threshold-

indexed local distribution has remained open.

Sui et al. (2026) study prediction-powered conditional functionals, and Testa et al. (2025) allow selective labels with decreasing overlap. For transported outcomes, Wang (2026) develops a framework for surrogate-assisted distributional inference. Local distribution estimation with fully observed outcomes has an established theory (Cattaneo et al., 2024). Generation-powered and multi-source extensions develop functional borrowing operators and confidence-region volume (Zhang and Li, 2026; Li et al., 2026). Anderson’s covariance-dominating operator improves Gaussian probabilities of centred convex symmetric sets (Anderson, 1955), and a fitted path may lack this ordering.

We develop *paired gain certification* through a common covariance comparison. The estimated object is the difference between Anchor and candidate simultaneous critical values, whose paired uncertainty vanishes at zero borrowing. Exploiting this cancellation yields an oracle bound proportional to the beneficial borrowing weight. An implementable multiplier calibration with independent numerical guards supports simultaneous inference after selection with nonunique optimal weights. A binary local experiment gives the reference-information scale for certifying weak improvements. The general learning-and-testing principle is established (Angelopoulos et al., 2021); our contribution is the paired critical-value geometry, its local information scaling, and its use in distributional inference.

Section 2 defines the estimator and algorithm, and Section 3 states the certification, inference and information results. Section 4 compares borrowing criteria and evaluates their extensions; Section 5 analyses dated air-monitoring collocations before the concluding discussion.

2 Local estimation and paired borrowing

2.1 The reference distribution and augmentation path

We observe $O = (W, R, RY)$, where W includes broadly observed X and S . Assume $R \perp Y \mid W$ and initially treat $\pi(W) = pe(W)$ as known, with e bounded above and

away from zero. Let $Z_y = \mathbf{1}(Y \leq y)$ and use the bounded nonnegative localisation kernel $K_h(X - x_0) = K\{(X - x_0)/h\}$ for the target neighbourhood. The kernel mass normalises the local CDF and determines its inferential scale:

$$\mu_h = \mathbb{E}K_h, \quad v_h = h^d, \quad f_h = \mu_h/v_h, \quad F_h(y) = \frac{\mathbb{E}(K_h Z_y)}{\mu_h}. \quad (1)$$

Localisation-bias control connects this distribution at resolution h to the point target $F_0(y \mid x_0)$.

Independent training produces bounded candidates $q_y(X)$ for Anchor and $m_y(W)$ for surrogate-assisted estimation. We define the common correction path by $D_y = m_y - q_y$ and $a_{\lambda,y} = q_y + \lambda D_y$, $0 \leq \lambda \leq 1$. The augmented signal separates Anchor from its weighted surrogate correction:

$$U_{\lambda,y} = q_y + \frac{R}{\pi}(Z_y - q_y) + \lambda \left(1 - \frac{R}{\pi}\right) D_y = U_{0,y} + \lambda B_y. \quad (2)$$

Conditional on training, $\mathbb{E}(U_{\lambda,y} \mid W) = \mathbb{E}(Z_y \mid W)$ and $\mathbb{E}(B_y \mid W) = 0$, so borrowing preserves the target.

On the threshold grid $\mathcal{Y}_J = \{y_1, \dots, y_J\}$, the stacked score combines Anchor fluctuation and correction; the linear map selects their weighted sum:

$$\Psi = \sqrt{p/v_h} K_h \begin{pmatrix} U_0 - F_h \\ B \end{pmatrix}, \quad \Omega = \mathbb{E}(\Psi \Psi^\top), \quad H_\lambda = (I_J, \lambda I_J). \quad (3)$$

This quadratic covariance path determines simultaneous uncertainty at every borrowing weight:

$$\Sigma_\lambda = H_\lambda \Omega H_\lambda^\top = \Omega_{00} + \lambda(\Omega_{01} + \Omega_{10}) + \lambda^2 \Omega_{11}. \quad (4)$$

For evaluation sample E, the kernel ratio defines the estimator and expresses its error

through the projected score:

$$\hat{F}_{E,\lambda} = \frac{\mathbb{P}_E K_h U_\lambda}{\mathbb{P}_E K_h}, \quad \sqrt{n_E p v_h}(\hat{F}_{E,\lambda} - F_h) = \frac{n_E^{-1/2} \sum_{i \in E} H_\lambda \Psi_i}{\hat{f}_{h,E}}. \quad (5)$$

Here $\hat{f}_{h,E} = \mathbb{P}_E K_h / v_h$; a zero denominator gives the full $[0, 1]$ distribution band.

2.2 The uncertainty criterion

For $G_\lambda \sim N(0, \Sigma_\lambda)$, the critical-value difference measures simultaneous precision gained over Anchor:

$$c_\Omega(\lambda) = \text{Quantile}_{1-\alpha} \|G_\lambda\|_\infty, \quad g_\Omega(\lambda) = c_\Omega(0) - c_\Omega(\lambda). \quad (6)$$

Positive gain shortens the Gaussian raw band at fixed resolution, sample size and confidence level. Rescaling both score blocks before forming Ω permits a fixed positive band shape common to all weights. Trace instead minimises $\text{tr}(\Sigma_\lambda)$, producing the distinction in the following example.

Proposition 1 (Separation of the criteria). *There is a valid distributional augmentation path for which the trace minimiser 25/41 reduces $\text{tr}(\Sigma_\lambda)$ from 0.340 to 0.318, but increases its 95% simultaneous critical value from 0.983562 to 0.996761.*

Take $S \sim \text{Bernoulli}(0.2)$, with $Y \mid S = 0$ uniform on $\{1, 2\}$ and $Y \mid S = 1$ uniform on $\{0, 2\}$. At thresholds $(0, 1)$, use $q = (0.1, 0.5)$, $m(S) = \{0.5S, 0.5 + 0.4(S - 0.2)\}$ and independent verification with $p = 0.1$. Figure 1 compares the criteria; the proof uses marginal Gaussian probability inequalities.

2.3 An implementable paired calibration

An independent gate sample G estimates Ω and accounts for Anchor centring. The centred gate residuals use $\hat{F} = \mathbb{P}_G K_h U_0 / \mathbb{P}_G K_h$ and $\hat{r}_i = (U_{0,i} - \hat{F}, B_i)^\top$. The gate estimates the

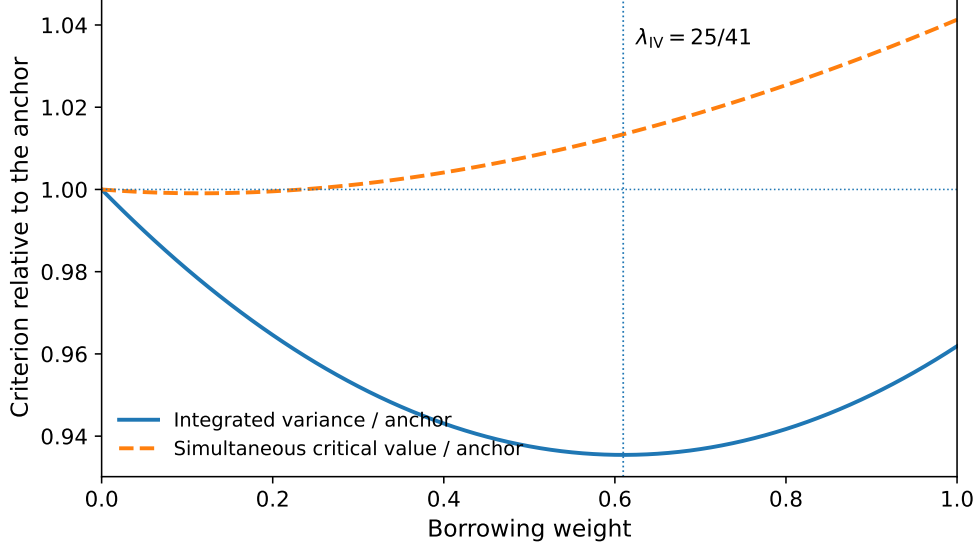


Figure 1: The exact two-threshold example separates the trace and simultaneous-band criteria. Both curves are relative to Anchor; at the trace optimum 25/41, the simultaneous critical value exceeds the anchor value. These are Gaussian calculations, not Monte Carlo estimates.

Alt text: The trace improves while the simultaneous criterion worsens at the trace optimum.

covariance, centring derivative and Anchor influence contribution:

$$\hat{\Omega} = \frac{p}{v_h} \mathbb{P}_G(K_h^2 \hat{r} \hat{r}^\top), \quad \hat{A} = \frac{p}{v_h} \mathbb{P}_G(K_h^2 \hat{r}), \quad \hat{\ell}_i = \frac{K_{h,i}(U_{0,i} - \hat{F})}{\mathbb{P}_G K_h}. \quad (7)$$

The covariance influence contribution explicitly incorporates the estimation of Anchor centring:

$$\hat{C}_i = \hat{\Psi}_i \hat{\Psi}_i^\top - \hat{\Omega} - E_0 \hat{\ell}_i \hat{A}^\top - \hat{A} \hat{\ell}_i^\top E_0^\top, \quad E_0 = \begin{pmatrix} I_J \\ 0 \end{pmatrix}. \quad (8)$$

Let Λ be a finite weight grid containing zero, with normalised gain gradients defined for $\lambda > 0$. These gradients convert covariance influence contributions into scores for calibrating gain uncertainty:

$$D_\lambda(\Omega) = \nabla_\Omega \{g_\Omega(\lambda)/\lambda\}, \quad \hat{s}_{i\lambda} = \langle D_\lambda(\hat{\Omega}), \hat{C}_i \rangle. \quad (9)$$

Centre the scores across observations; independent standard normal multipliers approxi-

mate $\max\{0, \max_{\lambda>0} n_G^{-1} \sum_i \xi_i \hat{s}_{i\lambda}\}$. Let $\hat{t}_\delta$ be an upper simulated quantile at probability $1 - \delta$. A nonnegative second-order cushion b_N and fresh Gaussian intervals $[\underline{c}_\lambda, \bar{c}_\lambda]$ protect the plug-in critical-value comparison. Paired selects the weight with the largest lower bound on simultaneous critical-value gain:

$$\hat{L}(0) = 0, \quad \hat{L}(\lambda) = \underline{c}_0 - \bar{c}_\lambda - \lambda(\hat{t}_\delta + b_N), \quad \hat{\lambda} \in \arg \max_{\lambda \in \Lambda} \hat{L}(\lambda). \quad (10)$$

Ties at zero select Anchor; comparisons share the same estimated joint covariance. The numerical studies use $b_N = \hat{c}_0 \log(1 + k_G)/k_G$, where k_G counts verified kernel-support observations, with k_G replaced by one when zero. Its asymptotic order under localisation is $\log N_G/N_G$, with local information index $N_G = n_G p h^d$.

Radial Gaussian integration evaluates the score gradient using the following probability representation. For a Cholesky factor L and a uniform direction V on the unit sphere in $\mathbb{R}^J$,

$$\Pr\{\|N(0, \Sigma)\|_\infty \leq t\} = \mathbb{E}_V F_{\chi_J}\{t / \|LV\|_\infty\}. \quad (11)$$

The Supplement derives the gradient and numerical conditions; fresh independent Gaussian draws provide the order-statistic intervals in (10). Their error probability enters the certificate separately from statistical calibration uncertainty.

3 Certification and simultaneous inference

3.1 A covariance-set benchmark and paired sensitivity

Conditional on training, let $\mathcal{C}_G$ be a nonempty compact set of positive semidefinite joint covariances with coverage at least $1 - \delta$. The paired lower bound compares critical values throughout this set and determines the ideal selector:

$$L_G(\lambda) = \inf_{Q \in \mathcal{C}_G} \{c_Q(0) - c_Q(\lambda)\}, \quad \lambda_G \in \arg \max_{\lambda \in \Lambda} L_G(\lambda). \quad (12)$$

Both critical values use the same Q , giving $L_G(0) = 0$ exactly for the common comparison. This benchmark separates the statistical certificate from the numerical approximation of its criterion.

Theorem 1 (Paired certification). *Let $\lambda^* \in \arg \max_{\lambda \in \Lambda} g_\Omega(\lambda)$. Then $\Pr\{g_\Omega(\lambda_G) < 0 \mid \mathcal{T}\} \leq \delta$. On $\{\Omega \in \mathcal{C}_G\}$,*

$$0 \leq c_\Omega(\lambda_G) - c_\Omega(\lambda^*) \leq \sup_{Q \in \mathcal{C}_G} |g_Q(\lambda^*) - g_\Omega(\lambda^*)|. \quad (13)$$

Neither uniqueness nor curvature of the minimiser is required.

The inequalities $g_\Omega(\lambda_G) \geq L_G(\lambda_G) \geq L_G(\lambda^*)$ and $L_G(\lambda_G) \geq 0$ reduce the oracle comparison to paired covariance sensitivity.

Theorem 2 (Anchor-adaptive sensitivity). *Fix $J, \alpha, \kappa > 0$ and $B < \infty$. Suppose Q, Q' belong to $\mathcal{K}_{\kappa, B} = \{Q \succeq 0 : \|Q\|_{\text{op}} \leq B, H_\lambda Q H_\lambda^\top \succeq \kappa I_J, 0 \leq \lambda \leq 1\}$. There is a finite $C = C(J, \alpha, \kappa, B)$ such that*

$$|g_Q(\lambda) - g_{Q'}(\lambda)| \leq C\lambda \|Q - Q'\|_{\text{op}}. \quad (14)$$

If $\mathcal{C}_G \subset \mathcal{K}_{\kappa, B}$ has radius ε_G about a common centre, the right side of (13) is at most $2C\lambda^\varepsilon_G$.*

The proof differentiates the Gaussian rectangle quantile twice with respect to covariance and uses the cancellation at $\lambda = 0$. The joint matrix may be singular, provided its output covariance is positive definite; this includes an exactly zero correction. The constants depend on dimension and conditioning; growing grids require the explicit dimension-dependent conditions in the Supplement.

3.2 Feasible calibration under scarce verification

Write $N_G = n_G p h^d$ and $N_E = n_E p h^d$ for the local information indices governing the implemented procedure. Within each triangular row, observations are independent copies

and the three sample roles are disjoint. Assume $\mu_h \asymp h^d$ with bounded kernels and candidate functions, together with verification bounds $0 < c_e \leq e \leq C_e < \infty$. Both information indices diverge, and output covariances remain in a compact positive definite set for fixed threshold and weight grids. The normalised nonzero covariance-influence scores satisfy the Lindeberg and covariance-consistency conditions in Supplementary Section S3. These conditions apply to the fitted path and accommodate misspecification of either regression.

Theorem 3 (Feasible paired calibration). *Under these conditions, the covariance estimator in (7) has the influence expansion (8), and*

$$\frac{g_{\hat{\Omega}}(\lambda) - g_{\Omega}(\lambda)}{\lambda} = \frac{1}{n_G} \sum_{i \in G} \langle D_{\lambda}(\Omega), C_i \rangle + O_P(N_G^{-1}) \quad (15)$$

uniformly over nonzero grid weights. Suppose $b_N \sqrt{N_G} \rightarrow 0$ and $N_G b_N \rightarrow \infty$, and the computed score gradients obey the consistency condition in Section S3. If the numerical critical-value intervals fail jointly with probability at most η_c , and the simulated multiplier quantile fails as an upper quantile with probability at most η_t , then (10) satisfies

$$\Pr\{g_{\Omega}(\hat{\lambda}) < 0 \mid T\} \leq \delta + \eta_c + \eta_t + o(1). \quad (16)$$

There is an event $\mathcal{A}_N$ with conditional probability at least $1 - \delta - \eta_c - \eta_t - o(1)$ and a nonnegative $R_N = O_P(N_G^{-1/2})$ such that, on $\mathcal{A}_N$, every grid oracle λ^ satisfies*

$$0 \leq c_{\Omega}(\hat{\lambda}) - c_{\Omega}(\lambda^*) \leq \lambda^*(R_N + \hat{t}_{\delta} + b_N) + e_{\text{mc}}(\lambda^*). \quad (17)$$

Here $\hat{t}_{\delta} = O_P(N_G^{-1/2})$, $e_{\text{mc}}(\lambda)$ sums the anchor and candidate numerical interval widths, and $e_{\text{mc}}(0) = 0$. The bound retains the certificate's failure probability; at an anchor oracle its zero regret is asserted on $\mathcal{A}_N$.

Theorem 1 gives finite-sample covariance-set certification, whereas Theorem 3 gives the implemented asymptotic multiplier guarantee. Exact finite-sample protection covers Monte

Carlo quantile errors; Section S2 supplies an observable Bernstein covariance ball for the benchmark. For $D = s_N \widetilde{D}$ with $s_N \rightarrow 0$, the score statement applies after normalisation by the nonzero path scale under the corresponding Lindeberg condition. The case $D = 0$ returns Anchor exactly under the implemented rule.

3.3 Coverage, whole distributions and rotating roles

Theorem 4 (Independent evaluation). *Let $\mathcal{L} \subseteq [0, 1]$ be a fixed finite set or the compact interval. Assume the evaluation score process admits a Gaussian approximation uniformly over $\lambda \in \mathcal{L}$, its covariance estimate is uniformly consistent, and its Gaussian maximum is uniformly continuous at the target quantiles. For any $\mathcal{L}$ -valued weight measurable with respect to the training and selection samples, the independently calibrated grid band has coverage at least $1 - \alpha - \eta_E - o(1)$, where η_E bounds its Monte Carlo quantile error. No convergence to a unique deterministic weight is needed.*

Conditional on training and gate data, the chosen weight is fixed in (5). Uniform Gaussian approximation and covariance consistency then establish the evaluation result. With a uniformly bounded local score envelope, Gaussian rectangle bounds apply when $\log^7(n_E J)/N_E \rightarrow 0$ (Chernozhukov et al., 2016). Continuous-index results use the local increment and entropy conditions in Section S4.

If grid limits (l_j, u_j) cover $F_h(y_j)$, define their whole-distribution envelopes by

$$l(y) = \max\{0, \max_{y_j \leq y} l_j\}, \quad u(y) = \min\{1, \min_{y_j \geq y} u_j\} \quad (18)$$

These envelopes cover the entire distribution, taking empty maxima and minima as zero and one. Inverting these envelopes gives simultaneous quantile bounds even at atoms, with the resulting width depending on grid resolution. A second-order localisation bias is negligible for the point target when $\sqrt{N_E} h^2 \rightarrow 0$; smooth positive density additionally gives the quantile-process derivative.

Calibrate each of three roles at error $\alpha/3$ and intersect their whole-distribution bands.

A union bound gives overall error at most $\alpha + o(1)$ with arbitrary dependence between those bands. Section S5 also gives valid convex combinations of endpoints, which can be wider.

3.4 How much reference information certifies a gain?

Consider equally likely $S \in \{-1, 1\}$ and the binary indicator $Z = \mathbf{1}(Y = 0)$, with $\Pr_\theta(Z = 1 \mid S) = 1/2 + \theta S$. Set $q = 1/2$, $D = dS$, $0 < d < 1/4$, and verify independently with probability p . The scaled augmentation variance along this path has the following form:

$$p \operatorname{Var}_\theta(U_\lambda) = \frac{1}{4} + (1 - p)(d^2 \lambda^2 - 2d\theta\lambda). \quad (19)$$

For $0 < \theta < d$, $\lambda^* = \theta/d$ and the optimal critical-value gain is $z_{1-\alpha/2}(1 - p)\theta^2 + O(\theta^4)$.

Theorem 5 (Local certification boundary). *If a positive-gain certificate is issued on an event A with $\Pr_0(A) \leq \delta$, then $\Pr_\theta(A) \leq \delta + \sqrt{2n_{GP}}|\theta|$. Localising the association to a box-kernel support of probability r_h replaces n_{GP} by $n_{GP}r_h$. Conversely, conditional on k local verified observations,*

$$\theta_L = \frac{1}{2k} \sum_{i=1}^k S_i(2Z_i - 1) - \sqrt{\frac{\log(1/\delta)}{2k}}, \quad \hat{\lambda} = \min\{1, (\theta_L)_+/d\} \quad (20)$$

has harmful-selection probability at most δ , and selects a positive weight with probability at least $1 - \beta$ if $\theta \geq \sqrt{\log(1/\delta)/(2k)} + \sqrt{\log(1/\beta)/(2k)}$.

The proof bounds observed-data divergence by $4n_{GP}\theta^2$ and uses a one-sided Hoeffding bound for the converse. In this submodel, the signal boundary is $N_G^{-1/2}$ and the oracle-gain boundary is N_G^{-1} . Retaining this resolution requires smaller numerical error and a weight mesh that resolves the local optimum.

4 Simulation study

4.1 Design, comparators and performance measures

The experiments separate three questions about borrowing: improvement in distribution estimation, reduction in simultaneous band width and support for a positive-gain certificate. Methods share observations, verification indicators and role assignments within each replication. Twelve cells use 500 replications each; Supplementary Table S1 reports the complete results. Table 1 compares all six methods at a common sample size and reference fraction.

We independently generate $X \sim U[-1, 1]$, $\varepsilon, \eta \sim N(0, 1)$ to form the reference outcome and surrogate:

$$Y = \mu(X) + \sigma(X)\varepsilon, \quad \mu(x) = 1.2x + 0.6x^2, \quad \sigma(x) = 0.8 + 0.25x^2, \quad S = \ell(X)Y + \tau\eta.$$

Alignment uses $\ell(x) = 1$ and $\tau = 0.4$; the weak-surrogate design changes τ to 1.6. The local-reversal design sets $\ell(x) = -1$ for $|x| \leq 0.4$ and $\ell(x) = 1$ elsewhere. The no-signal design takes $\ell(x) = 0$; the zero-path design sets the surrogate candidate equal to Anchor.

The target is $F_h(\cdot \mid 0)$ with an Epanechnikov kernel, $h = 0.35$, and seven thresholds $0.8\Phi^{-1}(u_j)$. The u_j are equally spaced from 0.1 to 0.9, with weight grid $\{0, 0.1, \dots, 1\}$. Sample sizes are 6,000 and 12,000, with reference fractions 0.05, 0.10 and 0.20. Bounded selective verification favours larger standardised surrogate readings; Section S9 gives its probability for comparison with random verification. Three equal roles fit Gaussian location–scale regressions, select borrowing and evaluate the distribution. Anchor uses mean features $(1, X, X^2)$; the surrogate mean additionally uses S .

Anchor and Full fix $\lambda = 0$ and 1, respectively, throughout each replication. Trace minimises the estimated covariance trace; Plug-in directly minimises the estimated simultaneous critical value. Projection estimates a matrix control variate from the same gate covariance, using ridge $0.05 \operatorname{tr}(\hat{\Omega}_{11})/J$. Its matrix mixes and reverses threshold corrections,

extending the scalar candidate class. Paired uses (10), with statistical selection error 0.049 and two numerical selection errors of 0.0005 each. Evaluation allocates 0.049 to Gaussian coverage and 0.001 to numerical calibration. Radial gradients use 8,192 directions, gate calibration uses 3,999 multiplier draws, and critical-value protection uses 32,768 independent Gaussian draws.

We report simultaneous coverage, within-replication band-width ratios to Anchor, selected weights and grid-integrated squared error (ISE),

$$\text{ISE}_J = J^{-1} \sum_{j=1}^J \{\hat{F}_h(y_j) - F_h(y_j)\}^2,$$

using the raw estimator on the inferential grid in every comparison. Harmful selection means $c_\Omega(\hat{\lambda}) > c_\Omega(0) + 10^{-4}$ for the population covariance of the fitted path. Deterministic quadrature evaluates the covariance and its Gaussian criteria for this assessment. Continuous summaries use independent-replication Monte Carlo standard errors (MCSE); coverage and harm frequencies receive binomial intervals.

4.2 Gains, reversal and the cost of certification

Across twelve cells, Paired coverage ranges from 0.944 to 0.968, with aligned mean width ratios 0.731–0.831. At $n = 6,000$ and $p = 0.10$ under random verification, grid ISE falls from 3.074×10^{-3} for Anchor to 1.567×10^{-3} for Paired, whose width ratio is 0.747. Full, Trace and Plug-in yield width ratios near 0.722 and grid ISE near 1.42×10^{-3} . Paired’s mean weight rises from 0.507 to 0.807 as p increases from 0.05 to 0.20; adoption rises from 0.862 to 1.000. Doubling the sample size at $p = 0.10$ raises mean weight from 0.718 to 0.822 and reduces the width ratio from 0.747 to 0.731. Under selective verification at $p = 0.05$, adoption is 0.752 and mean weight is 0.421.

Full has width ratio 0.954 and grid ISE 2.526×10^{-3} against 2.899×10^{-3} for Anchor. Paired chooses zero in all 500 replications under the stated budgets.

Full increases mean width by 11.4% and grid ISE from 3.080×10^{-3} to 3.909×10^{-3} ,

with harmful-selection frequency 0.994. Paired returns Anchor; Trace and Plug-in select positive borrowing in only 0.6% of replications. Projection has width ratio 0.805 and grid ISE 1.901×10^{-3} in this setting. Under no signal, Trace and Plug-in have harmful-selection frequencies 0.310 and 0.312 and mean Gaussian gains -0.000217 and -0.000232 , respectively. Their width ratios are near 1.000, and Paired records no harmful selections. Projection has width ratio 1.013 and mean gain -0.007431 ; every method reduces to Anchor in the exact zero-path cell.

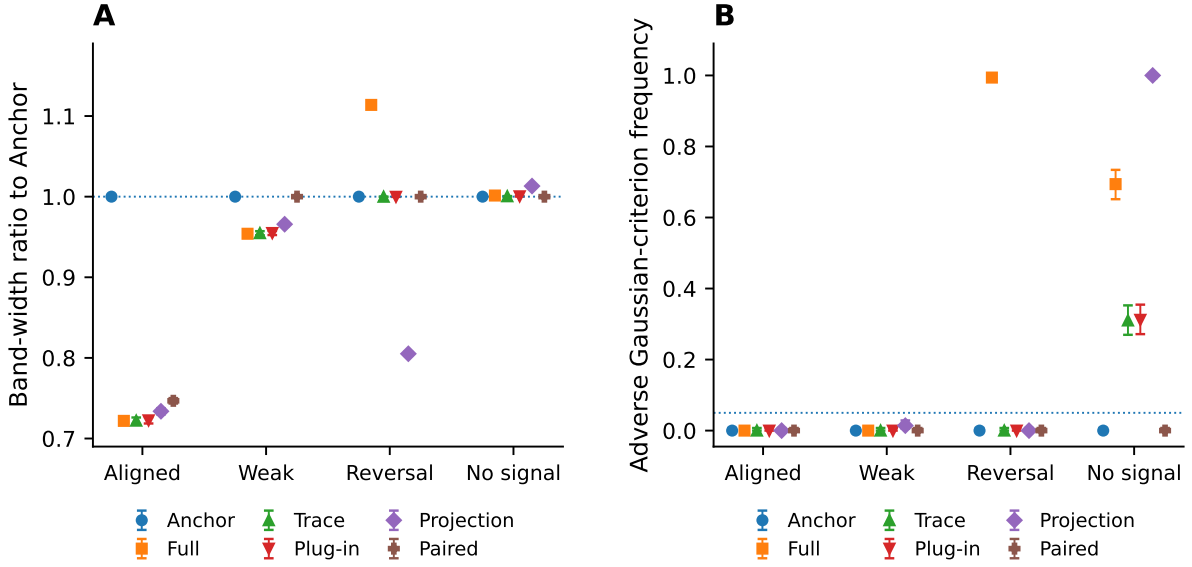


Figure 2: Six-method comparisons with $n = 6,000$ and $p = 0.10$ under random verification. (A) Mean within-replication band-width ratios; bars are ± 1.96 Monte Carlo standard errors. (B) Assessed harmful-selection frequencies, using critical-value tolerance 10^{-4} ; bars are exact binomial 95% intervals. Each mechanism has 500 replications. The dotted reference in (B) is 0.05.

Alt text: Projection benefits from reversal; Paired retains Anchor under weak and absent signals. Near-null harm frequencies can be high despite small width changes.

4.3 Pairing, dense thresholds and robustness

Six additional cells with 500 replications each compare Paired with Unpaired under a matched design. Unpaired separately bounds the Anchor critical value and the candidate-critical-value maximum; Section S3.5 establishes error control. Both selectors share fitted candidates, samples, weight meshes, numerical critical-value intervals and the total 0.05

selection-error budget. Unpaired adds two absolute-error bounds; Paired calibrates their difference using the gate scores. Under alignment at $p = 0.05, 0.10, 0.20$, Paired’s width ratios are 0.802, 0.747 and 0.769 against 1.000, 1.000 and 0.993 for Unpaired. Adoption probabilities are 0.858, 0.998, 1.000 for Paired and 0.000, 0.000, 0.032 for Unpaired. Both rules return Anchor under reversal and the no-signal correction. Paired coverage is 0.928–0.958, with low Anchor coverage in the weakest cell; Section S10 gives exact binomial intervals and all six-method comparisons.

The dense-grid experiment uses 21, 51 and 101 nested thresholds spanning probabilities 0.02–0.98 under alignment and selective local reversal. Each mechanism uses 200 datasets shared across resolutions and covariance enlargement $10^{-8}I_J$. Paired coverage is 0.935 to 0.965, with aligned width ratios 0.765, 0.782 and 0.798; it retains Anchor under reversal. The common 1,001-point grid evaluates monotone envelopes beyond the seven-coordinate primary experiment.

Seven further cells with 300 replications each examine standardised t_3 , skewed and mixture errors, a ten-category outcome, two-dimensional localisation, neighbour-distribution candidates and fitted logistic verification probabilities. The six known-design cells give Paired coverage 0.950–0.970 and width ratios 0.686–0.841. The learned-probability sensitivity gives coverage 0.963 and width ratio 0.768 under the nuisance-drift conditions in Section S8. Each generating distribution supplies the complete conditional CDF, including ordinal cutpoints.

4.4 Exact operating characteristics, rotation and numerical checks

Theorem 5 permits exact binary calculations independently of covariance quadrature and its numerical precision. With $p = 0.1$, correction amplitude $d = 0.2$ and certification error 0.05, we evaluate eight verified counts from 25 to 3,200. Eight associations run from -0.04 to 0.15; the 192 method–configuration entries are binomial probability sums. Figure 4 shows error control and adoption across the evaluated configurations. At zero association,

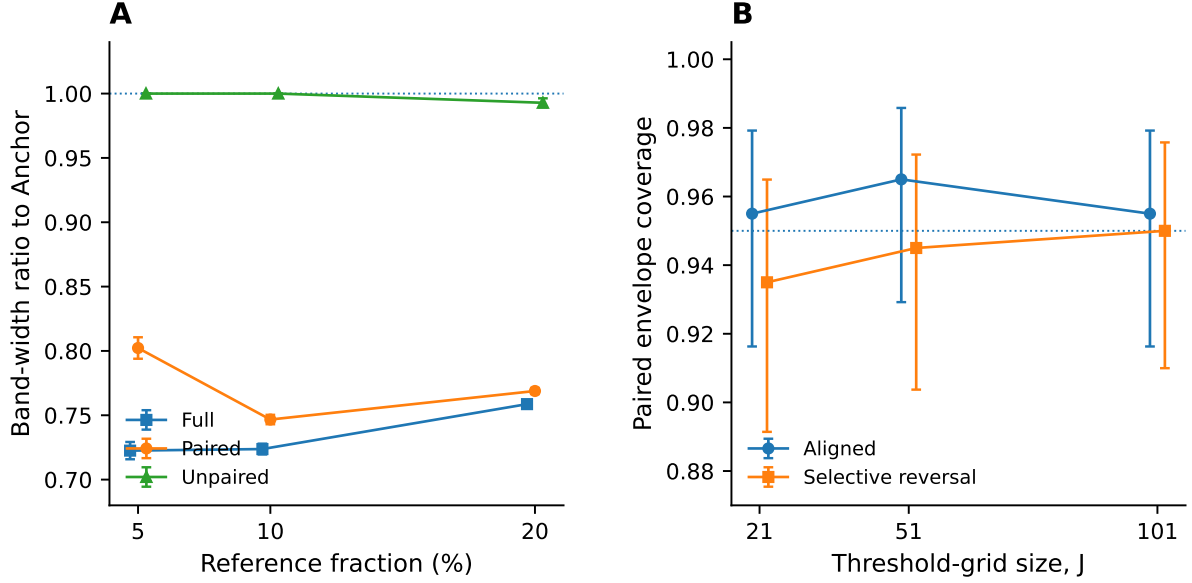


Figure 3: Pairing and dense-threshold validation. (A) Mean width ratios in the matched ablation, with ± 1.96 Monte Carlo standard errors of within-replication ratios, 500 replications per budget. Both certified rules use the same total error budget. (B) Paired monotone-envelope coverage on a common 1,001-point evaluation grid; bars are exact cell-wise binomial 95% intervals. The 200 datasets per mechanism are shared across threshold resolutions.

Alt text: Paired retains substantially more width improvement than the unpaired bound. Dense-grid envelope coverage is close to nominal, with visible Monte Carlo uncertainty.

Plug-in harmful-selection probability is 0.444 for $k = 50$ and 0.493 for $k = 3,200$. The exact binomial rule gives 0.032 and 0.047, and Hoeffding gives approximately 0.008 and 0.007. The largest error among both lower-bound rules over the grid is 0.0494. For $\theta = 0.04$, exact-binomial adoption rises from 0.100 at $k = 50$ to 0.727 at $k = 800$ and 0.998 at $k = 3,200$. Corresponding Hoeffding adoption probabilities are 0.031, 0.430 and 0.981 across those counts.

A separate 400-replication experiment uses 200 aligned and 200 reversed datasets with the dependence-valid intersection of three role-specific bands. Under alignment, Anchor and Paired coverage is 0.960 and 0.970, with mean widths 0.264 and 0.202. Under reversal, both methods have coverage 0.975 and mean width 0.263. Convex endpoint aggregation is wider and covers in every replication; Section S5 defines this separate experiment’s construction.

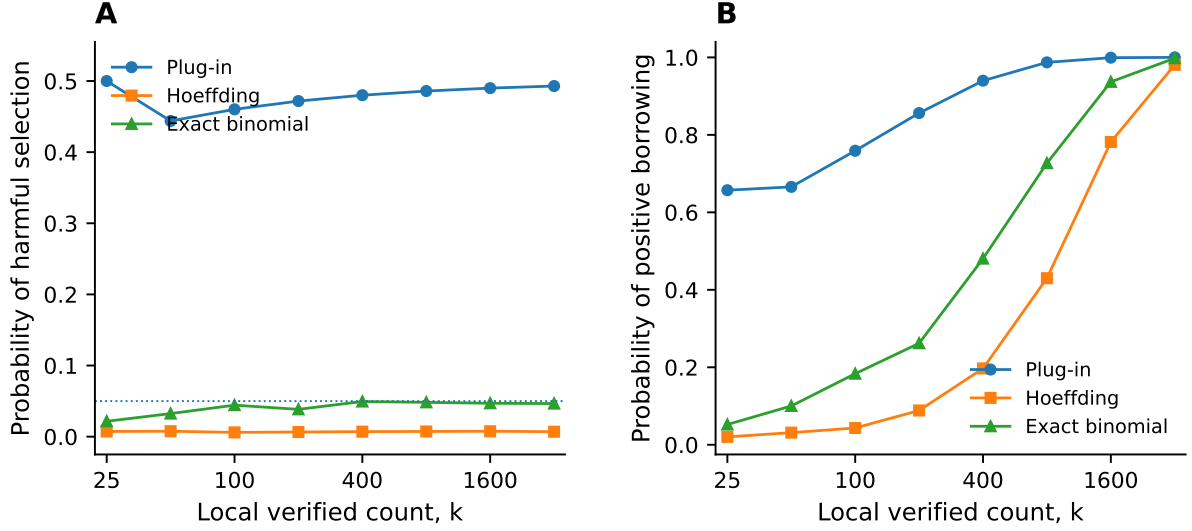


Figure 4: Exact binomial operating characteristics of the local submodel, with $p = 0.1$, $d = 0.2$ and certification error 0.05. (A) Harmful selection at $\theta = 0$. (B) Positive borrowing at $\theta = 0.04$. All values are exact binomial sums; the two lower-bound rules have distinct power despite controlling error.

Alt text: Plug-in harm approaches one half under the null; certified adoption increases with the verified count under a weak positive association.

All 6,000 primary replications finished successfully, and none required an information fallback. Across 60 checks, increasing radial directions from 8,192 to 32,768 preserved every Paired selection, with largest critical-value change 3.935×10^{-4} . The additional 6,300 design-grid replications use 5,500 distinct datasets because dense resolutions share samples. Refinement in 30 dense-grid cases changes the selected mesh point in 5 cases, with largest evaluation-width change 3.26% and estimated CDF change 0.0063. Section S10 retains all numerical comparisons, including every observed selection disagreement.

5 Humidity-specific air-monitor calibration

5.1 Scientific question and historical calibration

Can a mean-calibrated low-cost particle sensor recover the reference instrument’s exceedance frequencies, and which discrepancies can a limited reference sample establish? The United States Environmental Protection Agency (EPA) study collocated six particulate-

matter sensor types with reference instruments at seven United States sites (Barkjohn et al., 2025). Processed PurpleAir data provide 2,814 complete site-days from 22 July 2019 to 1 January 2021, covering 530 dates and 76 calendar weeks. Date recovery preserves the original observation identifiers and confirms unique site-date combinations. The target is the processed collocation series, retaining the source quality-control exclusions.

All 985 site-days in 2019 fit the historical calibration, which we freeze before subsequent distributional verification:

$$\tilde{S} = 3.098 + 0.486S - 1.127(X - 50)/20, \quad (21)$$

where S is uncorrected sensor concentration in micrograms per cubic metre and X is relative humidity in percent. Full-precision coefficients are stored in code; the remaining 1,829 site-days form the 2020–2021 audit frame. The same historical sample fits Anchor and surrogate distribution regressions. Their predictions and the calibration equation are frozen before reference sampling in the audit frame. The initial 985 reference measurements are additional to subsequent audit budgets.

Outcome regressions model $\log(1 + Y)$, where Y denotes reference concentration before transformation. Anchor uses humidity terms; the surrogate candidate adds sensor concentration and its interactions, with a fitted scale regression. Concentration thresholds are transformed inside these regressions, while scientific effects and intervals are reported on the original concentration scale. The archived whole-series leave-one-site-out diagnostic uses a different training design and is kept separate from this analysis.

5.2 The distributional target and complete-frame discrepancies

Kernel weights $K_i(r) = K\{(X_i - r)/10\}$ define the local contrast between reference and corrected-sensor distributions:

$$\Delta_H(t; r) = \frac{\sum_{i \in H} K_i(r) \{ \mathbf{1}(Y_i \leq t) - \mathbf{1}(\tilde{S}_i \leq t) \}}{\sum_{i \in H} K_i(r)}. \quad (22)$$

The contrast measures corrected-sensor minus reference exceedance frequency; positive values indicate overstatement in each humidity neighbourhood. The humidity targets $r = 30\%, 50\%, 70\%$ use the seven inferential concentration thresholds 5, 10, 15, 20, 25, 30, 40. The thresholds specify concentration levels used to assess distributional calibration. Nonzero kernel supports contain 379, 783 and 482 site-days before reference subsampling. The kernel effective sample sizes $(\sum K_i)^2 / \sum K_i^2$ are 315.889, 658.420 and 403.199, respectively.

Table 2 reports complete-frame calibration at 30% humidity: mean corrected-minus-reference concentration difference -0.191 and 0.90-quantile difference -1.612 . At concentration 15, corrected and reference exceedance frequencies are 6.068% and 9.307%, a difference of -3.239 percentage points. At 70% humidity, mean difference is -0.227 , 0.90-quantile difference is $+0.471$, and exceedance discrepancy at 15 is $+1.157$ percentage points.

In Figure 5, the discrepancy at 30% humidity changes from positive to negative, reaching 18.602 percentage points at concentration 5. Colorado receives 43.462% of kernel weight at 30% humidity and 0.793% at 70%. The target retains the recorded station-day mixture within each humidity neighbourhood.

5.3 Designed verification and finite-frame intervals

We independently randomise gate and evaluation masks conditional on the audit frame, giving design-based uncertainty for that frame. For a unit-specific union probability p_i , the two role probabilities are

$$\pi_{G,i} = p_i/2, \quad \pi_{E,i} = \frac{p_i - \pi_{G,i}}{1 - \pi_{G,i}}, \quad (23)$$

so $\Pr(R_{G,i} \vee R_{E,i} = 1) = p_i$ under independent gate and evaluation sampling. Random auditing uses $p_i = p$; selective auditing uses a bounded sensor-level score normalised to mean p , with budgets $p = 0.05, 0.10, 0.20$. Common uniforms nest each role's samples across budgets, and all five methods share the same masks. A unit selected by both roles

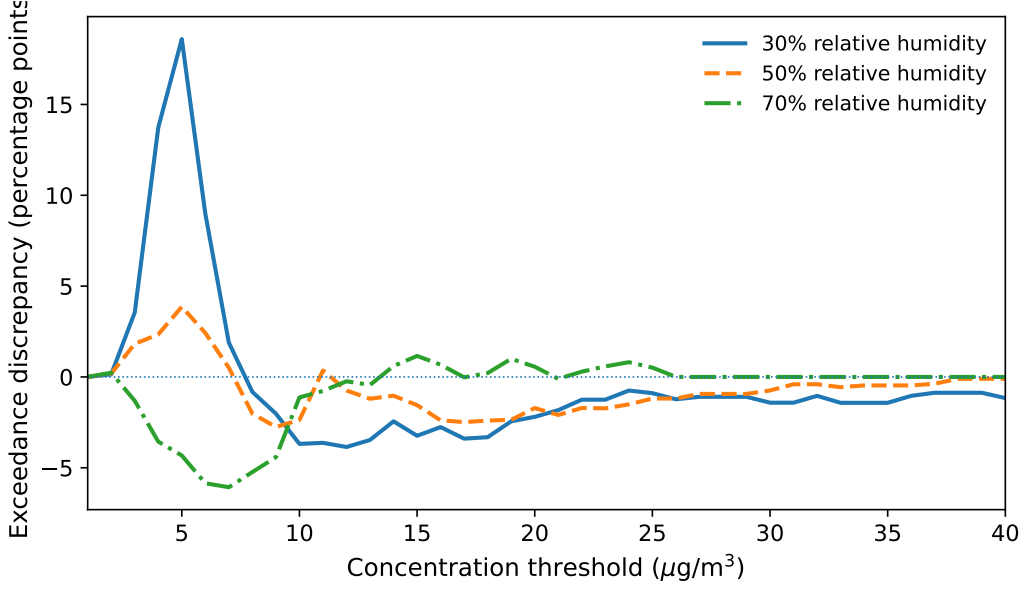


Figure 5: Complete-frame exceedance discrepancies after historical mean calibration. Positive values indicate overstatement by the corrected sensor. The curves use 40 integer concentration thresholds; inference uses the separately declared seven-threshold grid. These full-reference descriptions have no confidence limits.

Alt text: The discrepancy changes sign across concentration thresholds and differs among humidity neighbourhoods.

is measured once towards the unique reference budget. The masks remain independent conditional on the audit frame, including when their realised selections overlap.

Evaluation replaces reference indicators in (22) by $U_{\lambda,t}$ using R_E/π_E . With $w_i = K_i/\bar{K}$, $d_i = m_i - q_i$ and $r_i = (q_i - Z_i, d_i)^\top$, the joint matrix generates design covariance scaled by np :

$$\Omega_H = \frac{p}{n} \sum_{i \in H} w_i^2 (\pi_{E,i}^{-1} - 1) r_i r_i^\top. \quad (24)$$

The gate weights summands by $R_{G,i}/\pi_{G,i}$ and uses Bernoulli-design covariance-influence scores for paired calibration; Section S6 derives this calculation. Corrected-sensor indicators are known throughout the audit frame and cancel from its randomisation error.

Two hundred masks in each of six design-budget cells give 1,200 reference-sampling repetitions. Error budgets cover all 21 threshold-humidity combinations across the three humidity targets. The common Gaussian enlargement $10^{-8}I$ handles tail-grid singularities and preserves point estimates. Complete audit outcomes enter performance evaluation

only; neighbourhoods with fewer than three evaluation references receive full feasible distribution bands that remain included in coverage.

Supplementary Figure S1 shows the first retained random-audit mask at 20%, with 369 unique reference measurements. The gate/evaluation counts are 31/39, 72/104 and 48/66, and Paired weights are 0, 0.8 and 0. At 50% humidity, Paired width is 15.881 percentage points against 27.632 for Anchor and 14.831 for Full. Paired retains Anchor at 30% and 70%; the 70% interval at concentration 10 misses the complete-frame value.

Table 3 compares five methods, including the two unprotected adaptive selectors. Paired joint coverage is 0.935–0.970 across the six evaluated design–budget cells. The 0.935 result has Monte Carlo standard error 0.017, with exact binomial intervals retained for the comparisons. Under random auditing, Full has humidity-averaged width ratios 0.679, 0.667 and 0.660, against 0.985, 0.942 and 0.828 for Paired.

At the 5% random budget, the 30% humidity gate averages 9.93 references and Paired adoption is 1.5%. At the 20% budget, mean gate support rises to 38.46 and adoption to 40.5%. At 50% humidity, the counts are 19.67 and 77.37, with adoption rates 10.5% and 81.5%. Mean unique audit budgets are about 91, 183 and 365 measurements, in addition to the historical 985.

5.4 Reference budgets and upper-tail precision

A sign is resolved only when its simultaneous interval excludes zero in the correct direction. At 30% humidity and concentration 5, Paired resolution is 1.0%, 9.5% and 68.0% under 5%, 10% and 20% random audit budgets. Corresponding frequencies are 1.0%, 8.0% and 52.0% for Anchor and 32.0%, 55.0% and 89.0% for Full. At concentration 15 and the 20% budget, Paired resolution is 0%, 0.5% and 0% across humidity targets. These complete-frame differences are smaller than the discrepancy at concentration 5. All 126,000 method–mask–threshold records, including false-sign events, remain available for the evaluated grid.

A fixed follow-up evaluates 40%, 60%, 80% and 95% budgets through 400 nested

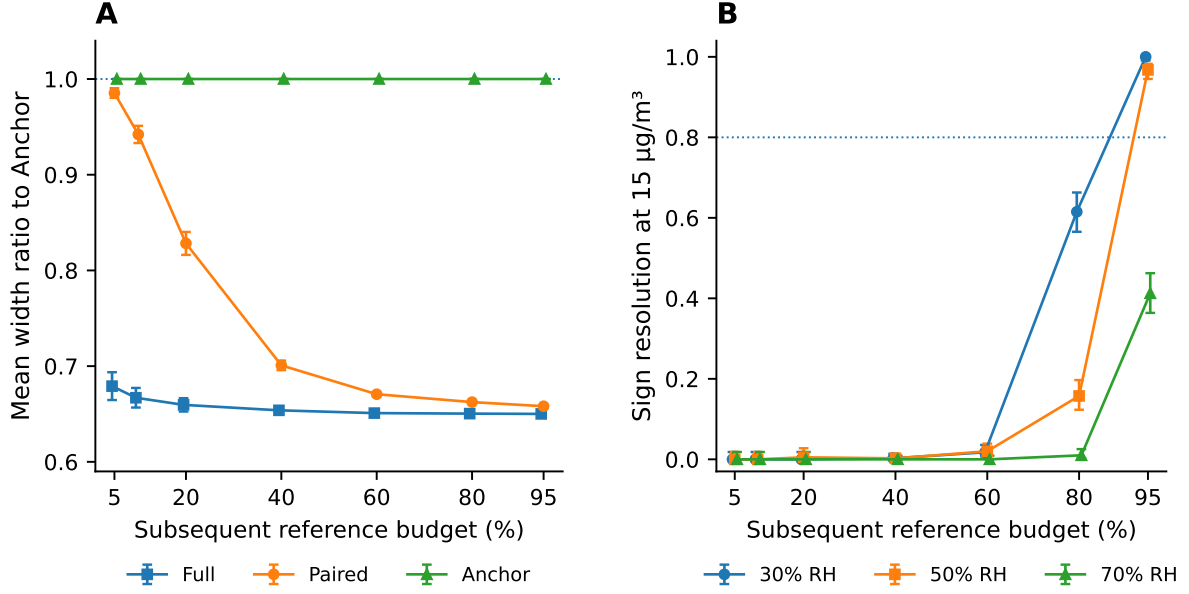


Figure 6: Reference-budget and upper-tail precision under random auditing. (A) Width ratios averaged across humidity within each mask, with ± 1.96 Monte Carlo standard errors. (B) Paired’s correct sign resolution at concentration 15, with exact binomial 95% intervals; the dotted line is 0.80. Budgets 5–20% use the original 200 masks and budgets 40–95% use 400 masks. Draw families are nested across budgets; historical training costs are additional.

Alt text: Increasing reference budgets allows smaller upper-tail differences to be resolved, but the 70-percent humidity discrepancy remains unresolved in many masks even at a 95-percent budget.

random-audit masks per budget. The follow-up retains all low-budget results and keeps historical regressions, thresholds, humidity targets and the selector unchanged. The 1,600 budget–mask combinations use four budgets sharing each of 400 independent uniform-draw families.

At concentration 15, Paired resolution probabilities at $p = 0.80$ are 0.615, 0.158 and 0.010 across humidity targets. At $p = 0.95$, they are 1.000, 0.968 and 0.412, the last with exact 95% Monte Carlo interval $[0.364, 0.462]$. Among evaluated subcensus budgets, 95% first exceeds 80% observed resolution at 30% and 50% humidity. No evaluated subcensus budget reaches that resolution at 70% humidity. A complete reference census determines the finite-frame contrast exactly as a separate endpoint. Joint coverage over all 21 combinations is 0.948–0.975; Figure 6 displays the complete budget trajectory. The 95% budget uses about 1,737 distinct audit references in addition to the historical 985.

The precision assessment retains all thresholds, incorrect-sign events and Monte Carlo intervals.

6 Discussion

Pairing candidate and Anchor critical values ties gain uncertainty to the amount borrowed. This cancellation makes the feasible multiplier calibration responsive to local reference information.

Under a common error budget, the matched ablation shows that a gain can be estimable as a difference while separate bounds remain too wide. Dense-grid and non-Gaussian experiments show that this mechanism extends beyond a short threshold vector.

The candidate class determines the available efficiency gains and their certification cost. Projection can exploit a reversed relationship outside the nonnegative scalar path, while Full can be more efficient under strong alignment. Paired controls whether the fitted correction is used along the proposed path. Insufficient gate information can leave a useful weak-surrogate correction uncertified under the stated confidence and numerical budgets.

Historical mean calibration and subsequent distributional verification answer different questions in the air-monitoring study. A modest audit establishes a large lower-concentration discrepancy; a one-percentage-point upper-tail difference can require nearly complete verification under the same simultaneous family. Designed reference sampling attributes this precision requirement to the audit frame and its recorded station-day mixture.

Data availability

The EPA analysis table, source-verified date mapping and original computational archive are available in the public repository <https://github.com/Stork343/surrogate-assisted-local-conditional-distribution-inference>, source commit 70c47c512ab485edf502f16914f9ff481cc6be88. The V3 reproduction package dated 29

September 2026 accompanying this manuscript contains the revised code, all generated results, input checksums and commands for rebuilding its tables and figures. The processed audit population and the historical calibration sample are identified separately.

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ChatGPT assisted the methodological revision, mathematical derivations, software development, numerical checks and drafting. Earlier computational development also used OpenAI Codex. These tools are not authors; responsibility for the submitted research, analysis and text rests with the authors.

Conflict of interest

The authors declare no conflicts of interest.

Supplementary material

The Supplement provides proofs, primitive and growing-grid conditions, numerical calibration details, rotating-role inference, the design-based audit extension and complete numerical results. The reproduction package provides executable code and the input and output data.

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Table 1: All six methods at a common sample size and reference fraction.

Mechanism	Method	Coverage	Grid ISE	Δ ISE (MCSE)	Gain
Aligned	Anchor	0.956	3.074	0.000 (0.000)	0.000
	Full	0.960	1.419	-1.656 (0.123)	180.739
	Trace	0.962	1.420	-1.654 (0.121)	180.373
	Plug-in	0.962	1.427	-1.647 (0.120)	179.937
	Projection	0.958	1.482	-1.592 (0.121)	172.550
	Paired	0.956	1.567	-1.508 (0.103)	162.655
Weak	Anchor	0.960	2.899	0.000 (0.000)	0.000
	Full	0.966	2.526	-0.373 (0.072)	31.131
	Trace	0.964	2.575	-0.324 (0.058)	29.995
	Plug-in	0.962	2.580	-0.318 (0.056)	29.746
	Projection	0.960	2.677	-0.222 (0.063)	23.600
	Paired	0.960	2.899	0.000 (0.000)	0.000
Reversal	Anchor	0.954	3.080	0.000 (0.000)	0.000
	Full	0.958	3.909	0.829 (0.063)	-72.220
	Trace	0.954	3.080	0.000 (0.000)	0.081
	Plug-in	0.954	3.080	0.000 (0.000)	0.081
	Projection	0.966	1.901	-1.178 (0.106)	125.115
	Paired	0.954	3.080	0.000 (0.000)	0.000
No signal	Anchor	0.968	3.000	0.000 (0.000)	0.000
	Full	0.968	3.006	0.007 (0.009)	-0.683
	Trace	0.966	3.003	0.003 (0.004)	-0.217
	Plug-in	0.966	3.005	0.005 (0.005)	-0.232
	Projection	0.958	3.105	0.105 (0.030)	-7.431
	Paired	0.968	3.000	0.000 (0.000)	0.000

Each row uses 500 replications with $n = 6,000$ and random verification at $p = 0.10$. Grid ISE, its mean paired difference from Anchor (Δ ISE) and mean Gaussian critical-value gain are multiplied by 10^3 . MCSE is the paired-difference Monte Carlo standard error. Negative Δ ISE and positive gain indicate improvement; Figure 2 reports widths and harmful-selection frequencies. Supplementary Table S1 contains all twelve cells and Monte Carlo uncertainties.

Table 2: Complete-frame calibration after historical training.

RH (%)	Local rows	Mean diff.	$Q_{.90}$ diff.	Ref. > 15	Sensor > 15
30	379	-0.191	-1.612	9.307	6.068
50	783	-0.380	-0.493	7.993	6.439
70	482	-0.227	0.471	7.038	8.194

Differences are corrected sensor minus reference in $\mu\text{g}/\text{m}^3$; exceedance frequencies are percentages. The frame has 1,829 site-days and the historical training sample 985. These are complete-frame descriptions.

Table 3: All five methods in the EPA reference-sampling experiment.

Design	p (%)	Anchor	Method			Paired
			Full	Trace	Plug-in	

(a) Joint coverage over all 21 threshold–humidity combinations						
Random	5	0.960	0.995	0.995	0.995	0.960
Random	10	0.955	0.970	0.965	0.970	0.960
Random	20	0.970	0.975	0.975	0.970	0.970
Selective	5	0.955	0.990	0.975	0.980	0.955
Selective	10	0.935	0.980	0.980	0.980	0.935
Selective	20	0.955	0.985	0.975	0.975	0.955
(b) Mean within-mask width ratio to Anchor						
Random	5	1.000	0.679	0.687	0.696	0.985
Random	10	1.000	0.667	0.669	0.674	0.942
Random	20	1.000	0.660	0.660	0.663	0.828
Selective	5	1.000	0.665	0.672	0.682	0.988
Selective	10	1.000	0.652	0.655	0.662	0.950
Selective	20	1.000	0.648	0.651	0.655	0.856

Each design–budget cell uses 200 masks. Widths are averaged across humidity within each mask. The average unique audit counts are 91.3, 182.8 and 364.8 under random auditing and 90.8, 183.5 and 364.1 under selective auditing. These counts exclude historical calibration. Projection is not evaluated in the EPA experiment.

Supplement to “Surrogate-assisted local conditional distribution inference with certified simultaneous gains”

Jian Hou, Tan Meng, and Maozai Tian

This Supplement supplies proofs and implementation details for the revised method. References to numbered theorems without an S prefix refer to the main manuscript. The independent-role population experiment and the fixed-frame Bernoulli audit have different covariance estimators; both are derived explicitly.

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S1 Geometry of paired critical values

S1.1 Notation and exact local identities

All expectations in Sections S1–S5 are conditional on the independent training sample. Dependence on the triangular row is suppressed. The threshold grid is $y_1 < \dots < y_J$. Set $Z_j = \mathbf{1}(Y \leq y_j)$, $D = m - q$, $a_\lambda = q + \lambda D$, and

$$U_\lambda = a_\lambda + R\pi^{-1}(Z - a_\lambda), \quad B = (1 - R/\pi)D.$$

With $R \perp Y \mid W$ and $\mathbb{E}(R \mid W) = \pi$, conditioning on (W, Y) gives $\mathbb{E}(U_\lambda \mid W, Y) = Z$ and $\mathbb{E}(B \mid W, Y) = 0$. Consequently $\mathbb{E}K_h(U_\lambda - F_h) = 0$. For n evaluation observations,

$$\begin{aligned} \sqrt{npv_h}(\hat{F}_\lambda - F_h) &= \frac{\sqrt{npv_h} n^{-1} \sum_i K_i(U_{\lambda,i} - F_h)}{\bar{K}} \\ &= \frac{n^{-1/2} \sum_i \sqrt{p/v_h} K_i(U_{\lambda,i} - F_h)}{\bar{K}/v_h}. \end{aligned} \tag{S1}$$

Thus denominator estimation need not be approximated for the coverage event: multiplying the estimated half-width by the same positive denominator cancels it exactly.

S1.2 Proof of the separation proposition

The four (S, Y) states $(0, 1), (0, 2), (1, 0), (1, 2)$ have probabilities 0.4, 0.4, 0.1, 0.1. The two candidate vectors are $(0, 0.42)$ and $(0.5, 0.82)$; each extends to a genuine cumulative distribution by taking value zero below zero and one at and above two. Enumerating these states and the independent verification indicator gives

$$\Sigma_\lambda = \begin{pmatrix} .09 - .072\lambda + .036\lambda^2 & .05 - .0288\lambda + .0288\lambda^2 \\ .05 - .0288\lambda + .0288\lambda^2 & .25 + .02304\lambda^2 \end{pmatrix}. \tag{S2}$$

Its trace is $.34 - .072\lambda + .05904\lambda^2$, with minimiser $25/41$. At zero borrowing, the union bound gives

$$\Pr(\|G_0\|_\infty > .99) \leq 2\bar{\Phi}(.99/.3) + 2\bar{\Phi}(.99/.5) = .04867037697 < .05.$$

At $25/41$, the second variance is $.25856632957$, so

$$\Pr(\|G_{25/41}\|_\infty \leq .99) \leq 2\Phi\{.99/\sqrt{.25856632957}\} - 1 = .94845626635 < .95.$$

Therefore $c_\Omega(0) < .99 < c_\Omega(25/41)$. Deterministic bivariate integration gives the values in the main text; numerical minimisation of the band criterion gives $\lambda = 0.114860$ and $c = 0.982639$. These latter values are not needed for the strict separation. An independent continuous X and a common kernel embed the example into a local conditional distribution, multiplying every covariance by the same positive constant.

S1.3 Gaussian-quantile differentiability

Let $q_\alpha(\Sigma)$ denote the $(1 - \alpha)$ quantile of $\|N(0, \Sigma)\|_\infty$. Define

$$H(t, \Sigma) = \int_{[-t, t]^J} \phi_\Sigma(z) dz, \quad t > 0.$$

On any compact subset of the positive definite cone, Gaussian density derivatives of every finite order are dominated by a polynomial times an integrable Gaussian. Differentiation under the integral is therefore valid. Substitution $z = tu$, $u \in [-1, 1]^J$, gives joint smoothness in t and Σ . The derivative in t is a sum of strictly positive face integrals, so $H_t(t, \Sigma) > 0$. Also $H(0, \Sigma) = 0$ and $H(t, \Sigma) \rightarrow 1$. The implicit function theorem applied to $H(q, \Sigma) = 1 - \alpha$ shows that q_α is infinitely continuously differentiable.

For a symmetric direction E , its first derivative is

$$Dq_\alpha(\Sigma)[E] = -\frac{\frac{1}{2} \int_{[-q, q]^J} \phi_\Sigma(z) \{z^\top \Sigma^{-1} E \Sigma^{-1} z - \text{tr}(\Sigma^{-1} E)\} dz}{H_t(q, \Sigma)}. \quad (\text{S3})$$

The quantile remains in a compact positive interval when the eigenvalues of Σ range over $[\kappa, 2B]$. Continuity and compactness give finite bounds M_r for its derivatives of orders $r = 1, 2, 3$. They depend on J, α, κ, B .

S1.4 Proofs of Theorems 1 and 2

On $\Omega \in \mathcal{C}_G$, the true gain dominates its lower bound at every weight. Since zero is a candidate, $L_G(\lambda_G) \geq L_G(0) = 0$. This proves the certificate. Moreover,

$$g_\Omega(\lambda_G) \geq L_G(\lambda_G) \geq L_G(\lambda^*) \geq g_\Omega(\lambda^*) - \sup_{Q \in \mathcal{C}_G} |g_Q(\lambda^*) - g_\Omega(\lambda^*)|.$$

Optimality of λ^* supplies the lower side of the regret inequality. A finite grid guarantees existence and measurability after deterministic tie-breaking.

For Theorem 2, write $f_\lambda(Q) = q_\alpha(H_\lambda Q H_\lambda^\top)$ and $\Sigma_\lambda = H_\lambda Q H_\lambda^\top$. Since $\|H_\lambda\|_{\text{op}}^2 \leq 2$,

$$\|H_\lambda E H_\lambda^\top - H_0 E H_0^\top\|_{\text{op}} \leq 3\lambda \|E\|_{\text{op}}, \quad \|\Sigma_\lambda - \Sigma_0\|_{\text{op}} \leq 3B\lambda.$$

The chain rule and the bounds M_1, M_2 yield

$$|D(f_0 - f_\lambda)(Q)[E]| \leq (3M_1 + 6BM_2)\lambda \|E\|_{\text{op}}.$$

The class $\mathcal{K}_{\kappa, B}$ is convex, so integrating along the segment between Q and Q' proves the paired Lipschitz bound. The confidence-set diameter is at most $2\varepsilon_G$, giving the stated regret. A third derivative bound similarly shows that the Hessian of $(f_0 - f_\lambda)/\lambda$ is uniformly bounded, including its continuous extension at zero. This supplies the second-order remainder used in Section S3.

A local numerical minimiser of $\inf_{Q \in \mathcal{C}_G} g_Q(\lambda)$ is generally an *upper* bound on the infimum. It cannot replace the lower bound in Theorem 1 without a global bound or a controlled approximation error. The implementation instead uses the influence calibration in the main algorithm.

S2 An observable covariance confidence set

The finite-sample benchmark can be instantiated without unknown second moments. We give one deliberately conservative construction for independent observations, a nonnegative compactly supported kernel and known verification probabilities. This construction is separate from the tighter asymptotic calibration used in the experiments.

Suppose $0 \leq K_h \leq K_{\max}$ and $\pi \geq pc_e$, with $c_e \leq 1$. Put $L = 2J + 1$, and define the raw vector and its second moment by

$$X_i = \sqrt{p/v_h}(K_i, K_i U_{0,i}^\top, K_i B_i^\top)^\top, \quad Q = \mathbb{E} X_i X_i^\top, \quad \hat{Q} = n^{-1} \sum_i X_i X_i^\top.$$

For $f \in [0, 1]^J$, let

$$A_f = \begin{pmatrix} -f & I_J & 0 \\ 0 & 0 & I_J \end{pmatrix}.$$

Then $\Omega = A_{F_h} Q A_{F_h}^\top$. Set $\hat{\mu} = \mathbb{P}_G K_h$ and let $\hat{f}$ be the coordinatewise clipping of $\mathbb{P}_G K_h U_0 / \hat{\mu}$ to $[0, 1]$. If $\hat{\mu} = 0$, take $\hat{f}_j = 1/2$. The benchmark estimate is $\hat{\Omega}_b = A_{\hat{f}} \hat{Q} A_{\hat{f}}^\top$.

Split δ into positive $\delta_K, \delta_Q, \delta_N, \delta_\mu$. For $s = \Pr(K_h > 0)$, let s^+ be its one-sided Clopper–Pearson upper confidence limit with error δ_K . Put

$$t_Q = \log(2L^2/\delta_Q), \quad t_N = \log(2J/\delta_N), \quad t_\mu = \log(2/\delta_\mu),$$

$$M_X = K_{\max}/(c_e \sqrt{pv_h}),$$

$$V_Q^+ = 2K_{\max}^4 s^+ / (pc_e^3 v_h^2),$$

$$V_N^+ = 2K_{\max}^2 s^+ / (pc_e),$$

$$V_\mu^+ = K_{\max}^2 s^+.$$

Use the radii

$$\begin{aligned} r_Q &= \sqrt{2V_Q^+ t_Q/n} + 4M_X^2 t_Q/(3n), \\ r_N &= \sqrt{2V_N^+ t_N/n} + 4K_{\max} t_N/(3np c_e), \\ r_\mu &= \sqrt{2V_\mu^+ t_\mu/n} + 2K_{\max} t_\mu/(3n). \end{aligned}$$

For $\hat{\mu} > 0$, define $\zeta = \min\{1, (r_N + r_\mu)/\hat{\mu}\}$; otherwise let $\zeta = 1$. Finally, set

$$\varepsilon_G = (J+1)Lr_Q + 2\sqrt{J(J+1)}\zeta\{\|\hat{Q}\|_{\text{op}} + Lr_Q\}. \quad (\text{S4})$$

Then

$$\Pr\{\|\hat{\Omega}_b - \Omega\|_{\text{op}} \leq \varepsilon_G \mid \mathbf{T}\} \geq 1 - \delta. \quad (\text{S5})$$

To prove this, note that $|U_0|, |B| \leq 1$ if $R = 0$, and both are at most $1/\pi$ if $R = 1$. Hence $\|X_i\|_\infty \leq M_X$ and

$$\mathbb{E}\|X_i\|_\infty^4 \leq 2K_{\max}^4 s/(pc_e^3 v_h^2).$$

Entrywise Bernstein inequalities and a union bound give $\|\hat{Q} - Q\|_{\max} \leq r_Q$. The analogous mean inequalities give numerator error r_N and denominator error r_μ . The event $s \leq s^+$ is intersected with the inequalities using the *true* support probability before substituting s^+ ; no independence between this confidence bound and the moments is assumed.

Since $\mathbb{E}K_h U_{0,j} = \mu_h F_h(y_j)$, the ratio error is at most $(r_N + r_\mu)/\hat{\mu}$, and clipping cannot increase it. The operator bounds

$$\|A_f\|_{\text{op}} \leq \sqrt{J+1}, \quad \|A_{\hat{f}} - A_{F_h}\|_{\text{op}} \leq \sqrt{J}\zeta, \quad \|\hat{Q} - Q\|_{\text{op}} \leq Lr_Q$$

now yield (S4) by expanding the difference of the two matrix products. The union error is at most the four allocated errors.

For fixed J , $\mu_h \asymp v_h$, $s \asymp v_h$, and $N_G = npv_h \rightarrow \infty$, this radius is $O_P\{\sqrt{\log(J/\delta)/N_G} + \log(J/\delta)/N_G\}$, with dimension-dependent constants. Intersect the positive semidefinite

ball with a known compact parameter class to obtain $\mathcal{C}_G$. If an imposed class produces an empty intersection, return the anchor. An intersection with $\mathcal{K}_{\kappa,B}$ requires the corresponding population condition; it cannot be used to discard an inconvenient covariance without that assumption.

S3 Feasible paired calibration and local information

S3.1 Primitive conditions and covariance expansion

The following conditions make the asymptotic assertions precise. For each row, the three samples consist of independent identically distributed observations with known $\pi = pe$, $c_e \leq e \leq C_e$ and $0 < p \leq 1$. The kernel satisfies $0 \leq K_h \leq K_{\max}$, $\Pr(K_h > 0) \asymp v_h$, $\mu_h \asymp v_h$, and $n_G p v_h \rightarrow \infty$. The trained candidates lie in $[0, 1]^J$. Conditional constants in these assumptions are uniform on training events with probability tending to one. Fixed h is allowed with v_h interpreted as a fixed positive constant.

Let $f = F_h$, $r = (U_0 - f, B)^\top$, $a_i = (p/v_h)K_i^2 r_i$, $A = \mathbb{E}a_i$, $E_0 = (I_J, 0)^\top$, and

$$\ell_i = K_i(U_{0,i} - f)/\mu_h, \quad C_i = \Psi_i \Psi_i^\top - \Omega - E_0 \ell_i A^\top - A \ell_i^\top E_0^\top.$$

Both ℓ_i and C_i have mean zero; A is generally nonzero. Boundedness and verification give, with constants depending on $J, c_e, K_{\max}$,

$$\mathbb{E}\|\Psi\|^2 = O(1), \quad \mathbb{E}\|\Psi\|^4 = O((pv_h)^{-1}), \quad \mathbb{E}\|\ell\|^2 = O((pv_h)^{-1}),$$

$$\|\hat{f} - f\| = O_P(N_G^{-1/2}), \quad \|\bar{a} - A\| = O_P(pN_G^{-1/2}), \quad \|A\| = O(p).$$

The last bound follows from $\mathbb{E}(U_0 \mid W) = m_0$, $\mathbb{E}(B \mid W) = 0$ and $\Pr(K_h > 0) = O(v_h)$.

Write $\Delta = \hat{f} - f$. The exact expansion of the centred second moment is

$$\begin{aligned}\hat{\Omega} - \Omega &= \mathbb{P}_G(\Psi\Psi^\top - \Omega) - E_0\Delta\bar{a}^\top - \bar{a}\Delta^\top E_0^\top \\ &\quad + (p/v_h)\mathbb{P}_G K_h^2 E_0 \Delta \Delta^\top E_0^\top.\end{aligned}\tag{S6}$$

The ratio expansion gives $\Delta = \mathbb{P}_G \ell + O_P(N_G^{-1})$ because the relative kernel-mean error is $O_P((n_G v_h)^{-1/2})$. Substitution yields

$$\hat{\Omega} - \Omega = \mathbb{P}_G C + O_P(N_G^{-1}), \quad \|\hat{\Omega} - \Omega\|_{\text{op}} = O_P(N_G^{-1/2}).\tag{S7}$$

The covariance-influence formula in the main text is the empirical version of this expansion. In particular, subtracting only $\hat{\Omega}$ from $\hat{\Psi}\hat{\Psi}^\top$ would miss the two centring derivatives.

S3.2 Score central limit theorem and multiplier calibration

For the $L_+ = |\Lambda \setminus \{0\}|$ nonzero weights, form the vector $s_i = (\langle D_\lambda(\Omega), C_i \rangle : \lambda > 0)$. For a non-collapsing path assume that

$$pv_h \mathbb{E}(s_i s_i^\top) \rightarrow V_s, \quad \mathbb{E}\{\|\sqrt{pv_h} s_i\|^2 \mathbf{1}(\|\sqrt{pv_h} s_i\| > \epsilon \sqrt{n_G})\} \rightarrow 0\tag{S8}$$

for every $\epsilon > 0$. The limiting Gaussian maximum, after adjoining zero, is assumed continuous at its positive $(1 - \delta)$ quantile. It suffices that at least one limiting score has positive variance and that this quantile is positive. The empirical score covariance is assumed consistent after multiplication by pv_h ; the bounded setup above gives this when the normalised score fourth moments are $O((pv_h)^{-1})$. No full-rank condition on V_s is needed.

Since L_+ is fixed, Cramér–Wold and the Lindeberg–Feller theorem give

$$\sqrt{N_G} \mathbb{P}_G s \Rightarrow N(0, V_s).$$

Conditional Gaussian multipliers have covariance $n_G^{-2} \sum_i \hat{s}_i \hat{s}_i^\top$. Consistency of that matrix, Gaussian continuity and the continuous mapping theorem therefore establish conditional weak convergence of their maximum. The difference caused by replacing $D_\lambda(\Omega)$ by $D_\lambda(\hat{\Omega})$ and C_i by its empirical version is negligible under (S7), compact covariance conditioning and the moment bounds.

For numerically evaluated gradients $\widetilde{D}_\lambda$, the required additional condition is

$$pv_h \mathbb{P}_G \left[\left\{ \langle \widetilde{D}_\lambda(\hat{\Omega}) - D_\lambda(\hat{\Omega}), \hat{C}_i \rangle \right\}^2 \right] = o_P(1) \quad (\text{S9})$$

uniformly over the finite weight grid. A sufficient condition for a fixed non-collapsing path is $\max_\lambda \|\widetilde{D}_\lambda - D_\lambda\|_{\text{op}} = o_P(1)$ and the displayed score moments. Radial integration satisfies this when its integration size increases and the derivative quadrature converges uniformly on the relevant compact covariance set. A fixed number of integration points is a numerical implementation choice, not an asymptotic theorem. The finite numerical checks in Section S9 assess this approximation separately from the exact order-statistic guard for critical values.

For $D = s_N \widetilde{D}$, $s_N > 0$, the same argument is applied to s_i/s_N . Require the analogues of (S8) and (S9) after this normalisation. If cancellation makes these normalised scores degenerate at another rate, their own nonzero scale must be used and checked. This condition explicitly distinguishes vanishing correction size from complete disappearance of information about its value. At $s_N = 0$, $B = 0$ and every critical value is identical; the algorithm returns zero.

S3.3 Proof of Theorem 3

Section S1 shows that $f_\lambda(Q) = g_Q(\lambda)/\lambda$ has a uniformly bounded Hessian for fixed J and covariances in the compact class. Taylor expansion and (S7) imply

$$f_\lambda(\hat{\Omega}) - f_\lambda(\Omega) = \langle D_\lambda(\Omega), \mathbb{P}_G C \rangle + O_P(N_G^{-1})$$

uniformly over nonzero weights. This proves the linear expansion. The maximum of the leading score errors is bounded by the conditional multiplier quantile with probability at least $1 - \delta - o(1)$. Its numerical upper-quantile construction costs an additional η_t . A cushion with $N_G b_N \rightarrow \infty$ absorbs the $O_P(N_G^{-1})$ remainder without affecting the first-order scale because $\sqrt{N_G} b_N \rightarrow 0$.

On the numerical interval event, $\underline{c}_0 - \bar{c}_\lambda \leq g_{\hat{\Omega}}(\lambda)$ for every weight. Intersect this event with the preceding statistical event. Then $\hat{L}(\lambda) \leq g_{\Omega}(\lambda)$ simultaneously, including the exact equality at zero. Maximisation gives $g_{\Omega}(\hat{\lambda}) \geq \hat{L}(\hat{\lambda}) \geq 0$. The union of the three failure probabilities gives Theorem 3.

For regret, on the same event,

$$g_{\Omega}(\hat{\lambda}) \geq \hat{L}(\hat{\lambda}) \geq \hat{L}(\lambda^*).$$

The loss between $g_{\Omega}(\lambda^*)$ and $\hat{L}(\lambda^*)$ is at most the absolute statistical gain-estimation error, $\lambda^*(\hat{t}_\delta + b_N)$ and the two numerical interval widths. Each statistical term has order $\lambda^* N_G^{-1/2}$. Define

$$R_N = \max_{\lambda \in \Lambda \setminus \{0\}} |g_{\hat{\Omega}}(\lambda) - g_{\Omega}(\lambda)| / \lambda.$$

The expansion gives $R_N = O_P(N_G^{-1/2})$. On the intersection of the statistical and numerical events the preceding inequality gives the explicit regret bound in the main theorem. At $\lambda^* = 0$, safety and optimality force zero regret on that event. Because its error probability need not vanish, this argument does not assert an unconditional zero regret at the anchor oracle.

The software sets $b_N = \hat{c}_0 \log(1 + k_G) / \max(k_G, 1)$. For a nondegenerate baseline, $\hat{c}_0$ stays bounded above and away from zero, and $k_G \asymp_P N_G$; the two cushion conditions follow. It is a diverging second-order cushion for an asymptotic statement, not a numerical value for the unknown finite-sample Taylor constant.

S3.4 Increasing thresholds and weights

The fixed-grid theorem does not hide a dimension-free covariance derivative. One transparent growing-grid formulation is the following. Let $J = J_N$, let the derivative bounds from S1 be $M_{r,N}$, and put $C_N = 3M_{1,N} + 6B_N M_{2,N}$. Suppose a covariance confidence set has error δ_N and radius ε_N . The exact finite-set certificate remains valid at every N , and

$$c_\Omega(\lambda_G) - \min_{\lambda \in \Lambda_N} c_\Omega(\lambda) \leq 2C_N \lambda_N^* \varepsilon_N$$

on its coverage event. Thus $C_N \varepsilon_N \rightarrow 0$ is an explicit sufficient condition for vanishing regret, without asserting that C_N is bounded.

For the feasible implementation require a simultaneous covariance expansion remainder $r_{\Omega,N}$, a slope Hessian bound T_N , and

$$\sup_{\lambda > 0} |\text{Taylor remainder in the slope}| \leq O_P\{C_N r_{\Omega,N} + T_N \varepsilon_N^2\} = o_P(b_N).$$

Require a Gaussian approximation for the normalised score-vector maximum and covariance consistency at the corresponding logarithmic dimension rate, together with numerical score error small on its approximation scale. These are verifiable sufficient conditions rather than a sharp universal growth rate. For bounded normalised scores with envelope $B_{s,N}$ and marginal variances bounded below, the high-dimensional rectangle criterion $B_{s,N}^2 \log^7(n_G L_N)/n_G \rightarrow 0$ is one sufficient route (Chernozhukov et al., 2016b). Output covariance singularity and low tail probabilities can deteriorate C_N, T_N and must be included in this check.

If a compact continuum of weights is approximated by a mesh of size Δ_λ , bounded $\partial_\lambda c_\Omega$ adds at most $C' \Delta_\lambda$ to oracle regret. Coverage of the whole distribution follows from the monotone envelope even for a fixed threshold grid. A claim of sharp continuum critical-value optimality would additionally need a quantitative modulus for the Gaussian threshold process; it is not used in the numerical studies.

S3.5 A matched unpaired certification comparator

The ablation holds the fitted path, gate and evaluation data, weight mesh, numerical critical-value intervals and total error budget fixed. It changes only how statistical uncertainty in the two critical values is controlled. Let

$$A_\lambda(\Omega) = \nabla_\Omega c_\Omega(\lambda), \quad v_{i\lambda} = \langle A_\lambda(\Omega), C_i \rangle.$$

Use centred empirical versions of these scores. Conditional Gaussian multipliers approximate separately the upper errors

$$\max\{0, n_G^{-1} \sum_i \xi_i \hat{v}_{i0}\}, \quad \max\{0, \max_{\lambda > 0} -n_G^{-1} \sum_i \xi_i \hat{v}_{i\lambda}\}.$$

Let t_0, t_1 be their upper simulated $(1 - \delta/2)$ quantiles. Each receives numerical quantile error $\eta_t/2$; the joint critical-value interval budget remains η_c . Define

$$L_U(0) = 0, \quad L_U(\lambda) = \underline{c}_0 - \bar{c}_\lambda - t_0 - t_1 - 2b_N \quad (\lambda > 0), \quad \hat{\lambda}_U \in \arg \max_{\lambda \in \Lambda} L_U(\lambda),$$

with ties at zero resolved in favour of the anchor. If the correction is identically zero, return the anchor directly.

Proposition S1 (Unpaired comparison under a matched error budget). *Under the fixed-grid conditions for the feasible paired calibration, with the corresponding nonzero absolute-score Lindeberg and covariance-consistency conditions, the unpaired selector satisfies*

$$\Pr\{g_\Omega(\hat{\lambda}_U) < 0 \mid \mathbf{T}\} \leq \delta + \eta_c + \eta_t + o(1).$$

Proof. The absolute critical-value expansions are $c_{\hat{\Omega}}(\lambda) - c_\Omega(\lambda) = P_G v_\lambda + O_P(N_G^{-1})$. The first multiplier controls $c_{\hat{\Omega}}(0) - c_\Omega(0)$; the second controls all $c_\Omega(\lambda) - c_{\hat{\Omega}}(\lambda)$, $\lambda > 0$. The two Taylor remainders are absorbed by $2b_N$ under the same cushion conditions as the paired theorem. Their simultaneous upper bound fails with probability at most $\delta + \eta_t + o(1)$ by

a union bound. On this event and the shared numerical interval event, $L_U(\lambda) \leq g_\Omega(\lambda)$ for every candidate. Maximising a family containing zero completes the proof. $\square$

The two absolute errors are computed from the same sample, but their separate upper bounds discard their statistical cancellation. Their penalty is generally of order $N_G^{-1/2}$ even as λ approaches zero, whereas the paired statistical penalty is multiplied by λ . This proposition gives both competitors the same advertised error budget. It does not assert that every lower-bound realisation or every chosen weight must be ordered. The ablation reports paired differences of realised performance.

S4 Post-selection bands and numerical calibration

S4.1 Proof of Theorem 4

Conditional on training, consider the evaluation scores $H_\lambda \Psi_i$. Their means are zero and their covariances are Σ_λ . Bounded candidates, known $\pi \geq pc_e$ and a compact kernel imply an envelope of order $(pv_h)^{-1/2}$. Their second moments are bounded. If output marginal variances are bounded below and $N_E \rightarrow \infty$, the fixed-dimensional Lindeberg central limit theorem holds. For a fixed finite grid of weights, it holds uniformly over that grid. For the compact interval, use the joint $2J$ -dimensional central limit theorem for Ψ_i and $H_\lambda \Psi_i$. Its continuous linear maps have uniformly bounded norms. A finite λ -net, stochastic equicontinuity from that linear representation, and the uniform lower output-variance bound give the asserted uniform approximation. The same argument applies to covariance estimation. This supplies the interval version of the main theorem rather than extending a finite-grid assumption to an arbitrary random weight.

Let $\hat{\Sigma}_{E,\lambda}$ be the evaluation second moment centred by its own estimated distribution. The expansion in S3, without the need to estimate the selection covariance influence function, gives uniform consistency. Gaussian continuity at the critical quantile yields conditional consistency of its plug-in critical value, or of a Gaussian evaluation multi-

plier. With an upper numerical order-statistic critical value, the additional probability of numerical under-calibration is at most η_E .

Now condition on both training and gate data. Evaluation observations retain their original conditional law and $\hat{\lambda}$ is fixed. Apply the uniform approximation to this realised weight and integrate over the conditioning samples. The exact ratio identity cancels the positive evaluation kernel denominator. No argument conditions simultaneously on all rotating gate choices. If the denominator is zero, the full distribution band is used and covers automatically. This proves the claimed lower coverage bound. A numerical upper critical value can be conservative at a fixed simulation size; nominal equality requires its numerical interval width to vanish.

A usual sufficient high-dimensional condition follows by appending positive and negative score coordinates, so the sup norm becomes a coordinate maximum. The bounded envelope has square of order $1/(pv_h)$; Gaussian rectangle approximation then follows from $\log^7(n_E J)/N_E \rightarrow 0$ under a lower marginal variance bound (Chernozhukov et al., 2016b). The cited rate is a sufficient condition, not a necessary local sample-size threshold.

For a continuous threshold process assume a compact $\mathcal{Y}$, locally bounded conditional density or an equivalent threshold-increment modulus, and fitted candidate classes of uniform VC type with bounded envelopes. Increments satisfy a local L^2 modulus tending to zero. The localised, inverse-probability weighted envelopes satisfy a triangular Lindeberg condition because $N_E \rightarrow \infty$. Finite-dimensional convergence follows as above; the entropy bound and the local increment modulus give asymptotic equicontinuity. Conditional multiplier equicontinuity follows from the same envelopes and entropy bound. These ingredients yield uniform process and multiplier convergence when the covariance kernels converge uniformly to a tight Gaussian limit (van der Vaart and Wellner, 1996; Chernozhukov et al., 2016a). The finite-grid envelope construction below avoids imposing this continuous-density condition on discrete outcomes.

S4.2 Monotone envelopes and quantile inversion

Suppose $l_j \leq F_h(y_j) \leq u_j$ for all j . For $y_j \leq y$, monotonicity gives $l_j \leq F_h(y)$. For $y_j \geq y$, it gives $F_h(y) \leq u_j$. Maximising the former and minimising the latter gives the whole-distribution envelope, with bounds 0 and 1 outside the extreme thresholds. Intersection with $[0, 1]$ preserves coverage. If the envelopes are inconsistent, replace them by $[0, 1]$; inconsistency can occur only outside the joint coverage event.

At probability τ , define extended-real bounds

$$q_L(\tau) = \sup\{y_j : u_j < \tau\}, \quad q_U(\tau) = \inf\{y_j : l_j \geq \tau\},$$

with empty bounds $-\infty, +\infty$. On the coverage event, $F_h(y_j) < \tau$ for every point in the first set and $F_h(y_j) \geq \tau$ for every point in the second. Hence $q_L(\tau) \leq Q_h(\tau) \leq q_U(\tau)$ simultaneously for all requested probabilities. This argument is valid for atoms and does not rely on an estimated density.

These monotone operations apply to a cumulative distribution. They are not applied directly to a difference of two distributions. A fixed known sensor distribution can be subtracted from the reference limits; a random superpopulation sensor distribution requires a joint process or a separate simultaneous error allocation.

S4.3 Exact order-statistic protection for Gaussian critical values

Let $T_1, \dots, T_B$ be independent draws from a continuous distribution F , conditional on all matrices and candidate weights. For $u \in (0, 1)$ and $q = F^{-1}(u)$, the number of draws below q is $K \sim \text{Binomial}(B, u)$. If integers l, r satisfy

$$\Pr(K < l) \leq \eta/2, \quad \Pr(K \geq r) \leq \eta/2,$$

then

$$\Pr\{T_{(l)} \leq q \leq T_{(r)} \mid F\} \geq 1 - \eta. \tag{S10}$$

Here $T_{(0)} = 0$ for nonnegative maxima and $T_{(B+1)} = +\infty$. The code takes l as the lower binomial quantile and r as one plus the upper binomial quantile. A one-sided upper critical value uses only the second inequality. Degenerate all-zero Gaussian maxima are handled exactly; repeated observations otherwise only make the bracketing conservative.

Apply (S10) to Gaussian sup norms generated from $\hat{\Sigma}_\lambda$, dividing η_c across the finite family of weights. Gaussian draws may be shared across weights: independence is needed across the B replicates for each marginal, not across weights. A union bound establishes simultaneous numerical intervals, even when the final weight is chosen using those intervals. The candidate grid must be fixed before these fresh draws. The multiplier maximum is another nonnegative random variable with a known conditional Gaussian sampling law; its upper order statistic is protected in the same way. The statistical bootstrap approximation and its Monte Carlo approximation are separate errors.

For a fixed well-conditioned Gaussian law with positive density at its target quantile, these interval widths have order $B^{-1/2}$ for fixed failure probability. Thus a gain of order N_G^{-1} requires $B/N_G^2 \rightarrow \infty$ for numerical error to be negligible unless an analytic or sharper paired numerical calculation is used. The scalar binary construction uses the exact normal quantile and has no such Monte Carlo floor.

S4.4 Radial integration and its derivative

For $Z \sim N(0, I_J)$ write $Z = RV$, with $R \sim \chi_J$ independent of the uniform spherical direction V . For a Cholesky factor L , the conditional event $\|LRV\|_\infty \leq t$ is $R \leq t/m(V)$, where $m(V) = \|LV\|_\infty$. This proves the radial representation. A spherical sample gives a smooth-in- t average of chi distribution functions; its quantile is found by bracketed root search.

At a direction with a unique maximising coordinate j^* , put $v = (LV)_{j^*}$ and $m = |v|$. Ties have spherical probability zero for a positive definite output covariance. The derivative

of $F_{\chi_J}(t/m)$ with respect to L is supported on row j^* and equals

$$-f_{\chi_J}(t/m)\frac{t}{m^2}\text{sign}(v)V^\top.$$

The derivative with respect to t is $f_{\chi_J}(t/m)/m$. Reverse differentiation through $\Sigma = LL^\top$, followed by division by the negative t derivative, gives the quantile gradient. The code validates this derivative against central finite differences and, in two dimensions, validates the quantile against deterministic rectangle integration.

The radial average is a Rao–Blackwellised distribution estimate, not an empirical distribution of independent scalar observations. An ordinary empirical-distribution Dvoretzky–Kiefer–Wolfowitz bound is therefore not invoked for it. Fresh direct Gaussian draws supply the rigorous numerical intervals. This distinction removes the unsupported numerical-confidence interpretation of the earlier exploratory quadrature check.

S4.5 Declared covariance enlargement

If output covariance is singular, one may specify a common ρI_J , $\rho > 0$, and define the criterion using $\Sigma_\lambda + \rho I_J$. All derivatives and paired gains then refer to this enlarged criterion. Adding an independent centred Gaussian vector decreases the probability of a centred convex symmetric set: its probability is maximised when the translation is zero (Anderson, 1955). Consequently the enlarged Gaussian critical value is conservative for the original Gaussian error law. The same ρ is used for every candidate, including the anchor. The EPA analysis sets $\rho = 10^{-8}$ in the stated covariance normalisation and does not claim unregularised critical-value optimality from this enlargement.

S5 Rotating roles and independent blocks

Let the observations be divided into disjoint folds I_1, I_2, I_3 . Rotation k trains on I_k , selects on I_{k+1} , and evaluates on I_{k+2} , with indices taken modulo three. For each rotation

separately, the other two folds are independent of its evaluation fold. This conditional argument gives a valid marginal band for that rotation. It does not justify conditioning jointly on all three fitted selections: a gate in one rotation is an evaluation fold in another.

Suppose $\mathcal{B}_k(y) = [l_k(y), u_k(y)]$ covers the common population target at all thresholds with probability at least $1 - \alpha_k - \epsilon_k$, for $k = 1, 2, 3$. On the intersection of the three coverage events,

$$\max_k l_k(y) \leq F_h(y) \leq \min_k u_k(y) \quad \text{for every } y. \quad (\text{S11})$$

The union bound gives probability at least $1 - \sum_k (\alpha_k + \epsilon_k)$, regardless of dependence among the bands. If the intersection is empty anywhere, the full $[0, 1]$ band replaces it. This replacement cannot reduce coverage. For fixed nonnegative ρ_k summing to one,

$$\sum_k \rho_k l_k(y) \leq F_h(y) \leq \sum_k \rho_k u_k(y)$$

holds on the same event. These are respectively the intersection and convex-aggregation procedures in the rotation experiment. The point estimate may still be the weighted mean of role-specific point estimates; the intersection interval need not be centred on that mean.

Separate gain certificates can also be allocated errors δ_k ; a union bound controls the probability of any role selecting a harmful gain by $\sum_k \delta_k + o(1)$. This is a family of role-specific covariance comparisons. Without an additional covariance argument it does not certify the Gaussian width of the *average estimator* formed from dependent rotations. No such pooled width certificate is claimed.

For the covariance-set version, let $U_k(\lambda) = \sup_{Q \in \mathcal{C}_k} c_Q(\lambda)$. On choosing a nonnegative paired lower bound, $c_Q(\hat{\lambda}_k) \leq c_Q(0)$ for every $Q \in \mathcal{C}_k$, hence $U_k(\hat{\lambda}_k) \leq U_k(0)$. The individual protected half-width is no larger than its equally protected anchor half-width. The convex sum preserves the half-width ordering; the intersection can move its centre, so a pointwise set inclusion is not implied merely by a width comparison.

For genuinely independent clusters, one can replace the individual score by a cluster sum and apply the same proofs with the number of clusters and cluster-level moment

conditions. Within-cluster dependence is then unrestricted subject to those conditions. Calendar weeks in a time series are not automatically independent clusters. With a known absolute-regularity bound and separated blocks, a coupling argument can transfer an independent-block result with an explicit total-variation remainder; without such a bound a numerical week-block bootstrap is a dependence sensitivity analysis, not a finite-sample theorem. The main EPA experiment conditions on its observed time series and instead uses independent verification randomisation, as proved next.

S6 Finite-frame verification of a distributional contrast

S6.1 Target and independent masks

Condition on an arbitrary finite frame $\{(W_i, Y_i) : i = 1, \dots, n\}$ and on historically trained candidates. No independence, stationarity or exchangeability of the frame values is assumed. Let $K_i \geq 0$, $\bar{K} > 0$, $w_i = K_i/\bar{K}$, $Z_i = (\mathbf{1}(Y_i \leq t_j) : j = 1, \dots, J)$, and $C_i^S = (\mathbf{1}(\tilde{S}_i \leq t_j) : j = 1, \dots, J)$. The target is

$$\Delta_H = n^{-1} \sum_i w_i (Z_i - C_i^S).$$

Generate independent Bernoulli indicators $R_{G,i}$ and $R_{E,i}$ with probabilities $\pi_{G,i}, \pi_{E,i}$, independent across frame indices and across roles. With $\pi_{G,i} = p_i/2$ and $\pi_{E,i} = (p_i - \pi_{G,i})/(1 - \pi_{G,i})$,

$$\Pr(R_{G,i} \vee R_{E,i} = 1) = \pi_{G,i} + \pi_{E,i} - \pi_{G,i}\pi_{E,i} = p_i.$$

Thus an overlapping observation counts only once towards the unique reference budget. Conditional on the frame, the two roles are independent despite possible overlap of realised labels. A historical sample used to learn the candidates must be accounted for separately;

it is not silently charged to either random audit mask.

S6.2 Exact evaluation error and covariance

For a gate-measurable weight, put $a_i = q_i + \lambda D_i$. The evaluation estimator is

$$\hat{\Delta}_\lambda = n^{-1} \sum_i w_i \{a_i + R_{E,i} \pi_{E,i}^{-1} (Z_i - a_i) - C_i^S\}.$$

Its error is exactly

$$\hat{\Delta}_\lambda - \Delta_H = n^{-1} \sum_i w_i (1 - R_{E,i} / \pi_{E,i}) (a_i - Z_i). \quad (\text{S12})$$

The known sensor part cancels. Hence, for any positive scaling constant p_* ,

$$np_* \text{Var}_E(\hat{\Delta}_\lambda \mid H, G) = H_\lambda \Omega_H H_\lambda^\top,$$

$$\Omega_H = n^{-1} \sum_i v_i v_i^\top, \quad v_i = \sqrt{p_*(\pi_{E,i}^{-1} - 1)} w_i \begin{pmatrix} q_i - Z_i \\ D_i \end{pmatrix}. \quad (\text{S13})$$

The code takes $p_* = n^{-1} \sum_i \pi_{E,i}$. This positive common normalisation cancels from selection and yields half-width $c/\sqrt{np_*}$. Kernel weights are already normalised, so no additional factor of h or estimated density is applied in the fixed-frame code.

The gate Horvitz–Thompson estimate is

$$\hat{\Omega}_H = n^{-1} \sum_i \frac{R_{G,i}}{\pi_{G,i}} v_i v_i^\top.$$

Every unobserved Z_i is absent from this sum. This estimate is positive semidefinite and unbiased. Its exact fluctuation is $n^{-1} \sum_i (R_{G,i} / \pi_{G,i} - 1) v_i v_i^\top$; no distribution-centring term is needed because the frame weights and target error representation are fixed.

For a symmetric slope gradient $D_\lambda(\Omega_H)$, put $a_{i\lambda} = v_i^\top D_\lambda(\Omega_H) v_i$. The covariance of the

linearised slope error is

$$V_{\lambda\gamma} = n^{-2} \sum_i \frac{1 - \pi_{G,i}}{\pi_{G,i}} a_{i\lambda} a_{i\gamma}.$$

An unbiased observed-data version replaces each summand by $R_{G,i}(1 - \pi_{G,i})a_{i\lambda}a_{i\gamma}/\pi_{G,i}^2$. The program forms scores $\sqrt{1 - \pi_{G,i}} a_{i\lambda}/\pi_{G,i}$ only on observed gate indices. Their Gram matrix divided by n^2 is precisely this variance estimate. Replacing the gradient by its consistent gate estimate gives the feasible variance used by the multiplier.

For the selected a_i , an independent evaluation covariance estimate is

$$\hat{\Sigma}_{E,\lambda} = \frac{p_*}{n} \sum_i \frac{R_{E,i}}{\pi_{E,i}} w_i^2 (\pi_{E,i}^{-1} - 1) (a_i - Z_i)(a_i - Z_i)^\top.$$

It is also unbiased conditional on the gate and frame. This differs from a superpopulation sample covariance because finite-frame randomisation has zero-mean errors at each unit, rather than a common unknown distribution mean.

S6.3 Asymptotic justification for frame sequences

Consider a sequence of deterministic frames. Require $\max_i p_i < 1$, probabilities comparable to their frame average, no single weighted observation dominating the selection or evaluation variances, consistency of the displayed covariance estimators, and output covariance matrices bounded above and below in the declared normalisation (or a fixed common enlargement). More explicitly, for every unit vector u in the relevant finite score dimension, the Bernoulli triangular array must satisfy

$$\frac{\sum_i \mathbb{E}\{(u^\top X_{i,n})^2 \mathbf{1}(|u^\top X_{i,n}| > \epsilon s_{u,n})\}}{s_{u,n}^2} \rightarrow 0, \quad s_{u,n}^2 = \sum_i \mathbb{E}(u^\top X_{i,n})^2,$$

for each $\epsilon > 0$, for both the covariance-slope and evaluation-error arrays. This condition is directly implied by bounded normalised contributions divided by their total standard deviations tending to zero. Require the corresponding empirical variance ratios to converge to one on nonzero directions.

Independent non-identically distributed Lindeberg–Feller convergence then gives the score and evaluation Gaussian limits. The quantile differentiability and Taylor argument from S3 apply to the gate moment estimate. Independent evaluation is valid conditional on the frame and gate. Numerical interval guards and error budgets are unchanged. This proves the design analogue of the feasible certificate and coverage theorem. Since these are frame-sequence asymptotics, very small local reference counts can still under-cover; the data experiment reports their observed coverage and does not relabel the theorem as an exact randomisation interval.

S6.4 EPA implementation and scientific interpretation

The historical calibration uses all 2019 rows and predicts later rows through $Y = \beta_0 + \beta_1 S + \beta_2(X - 50)/20$. Historical candidate regressions use log concentration. The anchor mean features are $1, x, x^2$, $x = (X - 50)/20$; the surrogate features add s, s^2, xs , $s = \log(1 + S)$. Each candidate variance model regresses $\log\{\max(\hat{\varepsilon}^2, 10^{-4})\}$ on the same features, adds 1.270362845 for the normal log-square expectation, and clips the resulting standard deviation to $[0.12, 2.5]$. All choices are fixed for this reanalysis and are recorded in `epa_protocol.json`.

The selective probability score is $1 + 0.4 \tanh\{(s - \bar{s})/\text{sd}(s)\}$, divided by its frame mean and multiplied by the requested p . This uses only broadly observed quantities. The three humidity-specific criteria are calibrated separately with a Bonferroni division of the total error, giving joint control for the declared three targets. Additional thresholds or humidity targets are not added after inspecting which confidence intervals exclude zero.

Complete audit outcomes compute the fixed-frame benchmark, diagnostic curves and criterion performance. They never fit a candidate or select a weight after masking. Code tests change unobserved outcomes and verify that the selected weights and corresponding estimates remain unchanged. Coverage describes repeated random verification of this processed population, not repeated sampling of United States sites or years. The station-weight mixture is retained; none of the between-humidity comparisons is interpreted as a

causal humidity effect.

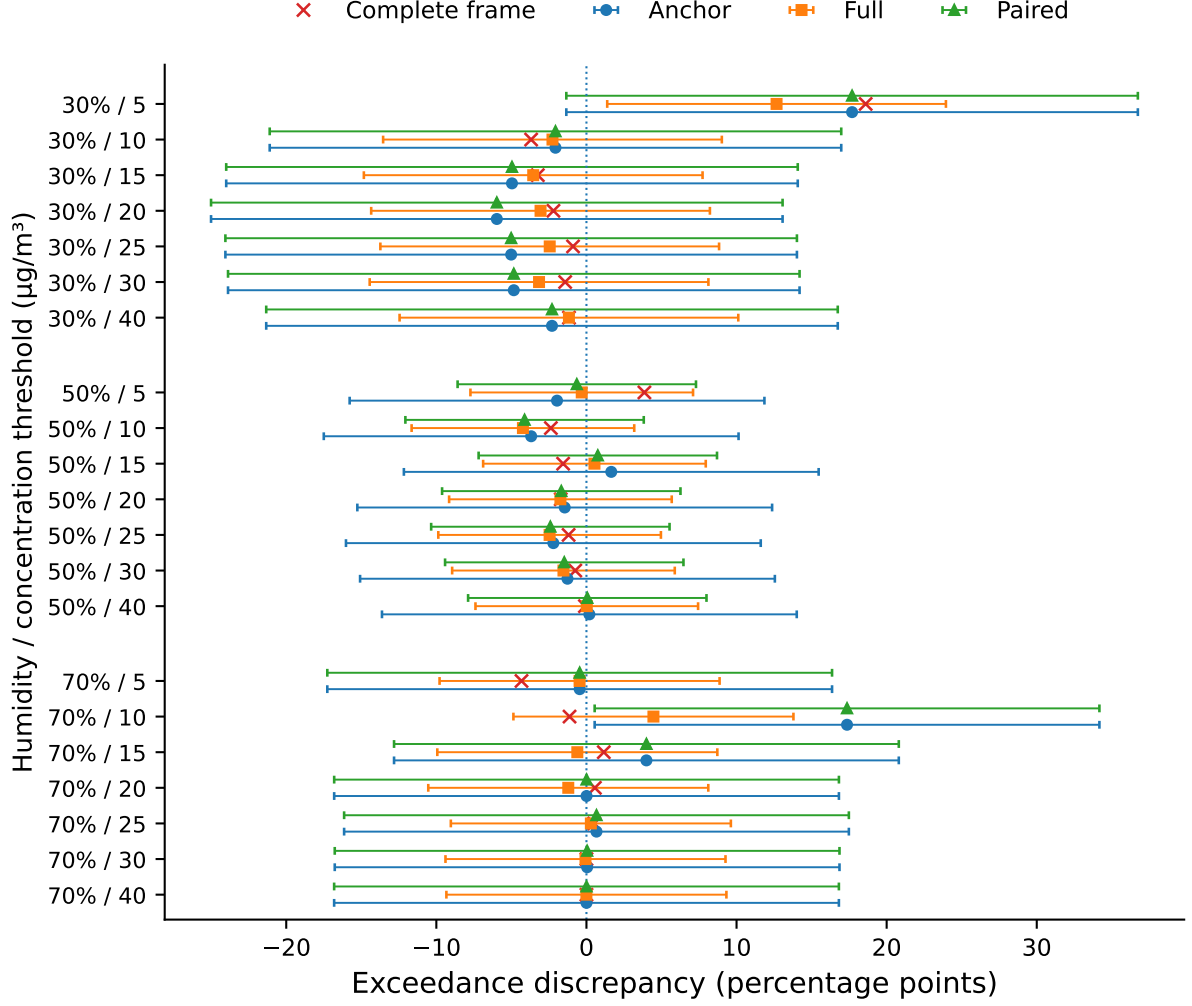


Figure S1: Actual simultaneous limits in the first retained random-audit mask at the 20% budget (369 unique references). All three humidity targets and seven thresholds are shown. Horizontal bars are limits for each method; crosses mark complete-frame values. Paired weights are $(0, 0.8, 0)$ at $(30, 50, 70)\%$ humidity. Repeated-mask coverage is reported in Table 3 of the main manuscript.

Alt text: Paired coincides with Anchor at 30 and 70 percent humidity and is narrower at 50 percent. One 70-percent interval misses its complete-frame value.

S7 Weak-signal information boundary

Let $S \in \{-1, 1\}$ equally likely, $\Pr_\theta(Z = 1 \mid S) = 1/2 + \theta S$, $|\theta| < 1/4$, and $R \sim \text{Bernoulli}(p)$ independently. The augmentation path is $a_\lambda = 1/2 + \lambda dS$, $0 < d < 1/4$. Conditioning on

(S, Z) gives $\mathbb{E}U_\lambda = 1/2$.

Write $V = Z - 1/2$, so $\mathbb{E}V^2 = 1/4$ and $\mathbb{E}SV = \theta$. The centred augmented signal is $RV/p + (1 - R/p)\lambda dS$. Expanding its square gives

$$p \operatorname{Var}(U_\lambda) = \frac{1}{4} + (1 - p)(d^2\lambda^2 - 2d\theta\lambda).$$

For $p < 1$ and $\theta \leq 0$, every positive weight increases variance. For $0 < \theta < d$, the optimum is $\lambda^* = \theta/d$ and

$$g_\theta^* = z_{1-\alpha/2} \left\{ \frac{1}{2} - \sqrt{\frac{1}{4} - (1 - p)\theta^2} \right\} = z_{1-\alpha/2}(1 - p)\theta^2 + O(\theta^4).$$

The expansion is uniform for p bounded away from one. At $p = 1$ all augmentation corrections are zero and no gain exists.

The observed laws of (S, R) coincide under θ and zero. Conditional on $R = 1$, the Bernoulli relative entropy to success probability $1/2$ is at most its chi-squared divergence $4\theta^2$. Thus

$$\operatorname{KL}(P_\theta^{n_G}, P_0^{n_G}) \leq 4n_G p \theta^2.$$

Pinsker's inequality gives total variation at most $\sqrt{2n_G p}|\theta|$. For any event A with $P_0(A) \leq \delta$,

$$P_\theta(A) \leq \delta + \sqrt{2n_G p}|\theta|.$$

To obtain a local model, take an independent X and let the association be θ only within the box-kernel neighbourhood. If its probability is r_h , each observation contributes at most $4pr_h\theta^2$ to divergence. The effective information is therefore $n_G pr_h \asymp n_G p h^d$. All unverified covariate and surrogate distributions agree across the alternatives.

For the converse certificate, conditional on k verified observations, each $S_i(2Z_i - 1)/2$ lies in $[-1/2, 1/2]$ with mean θ . Hoeffding's one-sided inequality implies $\Pr(\theta_L > \theta) \leq \delta$. On

the complementary event, a positive chosen weight satisfies $0 < d\hat{\lambda} \leq \theta_L \leq \theta$. Consequently

$$p \operatorname{Var}(U_{\hat{\lambda}}) - p \operatorname{Var}(U_0) = (1 - p)d\hat{\lambda}(d\hat{\lambda} - 2\theta) \leq -(1 - p)(d\hat{\lambda})^2 < 0.$$

Another one-sided Hoeffding bound gives adoption probability at least $1 - \beta$ under the signal inequality in the main theorem. Intersecting adoption with the valid lower-bound event gives probability at least $1 - \beta - \delta$ of positive, beneficial borrowing. For $k = 0$ use zero borrowing. This establishes matching fixed-submodel signal orders $k^{-1/2}$ and corresponding squared critical-gain order k^{-1} . Constants and multiplicity change with the requested confidence levels.

The target $P_\theta(Z = 1) = 1/2$ is constant in this submodel. Its role is to distinguish information needed to certify a variance reduction from information needed to estimate an otherwise known target; it is not a minimax lower bound for estimating F_h across a general nonparametric model. The exact binomial enumeration in the reproduction code computes the certificate's adoption and harmful-selection probabilities without simulation error.

S8 Reference efficiency and extensions of the original framework

S8.1 Canonical gradient and covariance excess

For fixed h and verification law, let $m_{0,y}(W) = \mathbb{E}(Z_y \mid W)$ and $F_h(y) = \mathbb{E}(K_h Z_y) / \mu_h$. The canonical gradient is

$$\phi_{h,y}^{\text{eff}}(O) = \frac{K_h}{\mu_h} \left\{ m_{0,y}(W) + \frac{R}{\pi} (Z_y - m_{0,y}(W)) - F_h(y) \right\}. \quad (\text{S14})$$

To verify the gradient identity, write an observed-data score as $s_W(W) + R s_Y(Y, W)$, where $\mathbb{E} s_W = 0$ and $\mathbb{E}(s_Y | W) = 0$. The pathwise derivative of F_h is

$$\mu_h^{-1} \mathbb{E}\{K_h(Z_y - F_h)(s_W + s_Y)\}.$$

Taking the inner product of (S14) with the observed score yields exactly this derivative, by iterated expectation and $\mathbb{E} R | W = \pi$. The displayed gradient lies in the observed tangent space: its first component is a mean-zero function of W , and its second is R times a conditionally mean-zero residual. If the verification law is unknown, its additional tangent scores have form $(R - \pi)b(W)$; their inner products with (S14) vanish. Thus the same efficiency bound applies. This is the standard missing-data tangent-space argument specialised to the local distribution (Tsiatis, 2006; Robins et al., 1994).

For any fixed augmentation a , write $\delta_y^a = a_y - m_{0,y}$ and $C_{yz}(W) = \text{Cov}(Z_y, Z_z | W)$. Then

$$U_y(a) = m_{0,y} + \frac{R}{\pi}(Z_y - m_{0,y}) + (1 - R/\pi)\delta_y^a.$$

Conditional cross-covariances between the two residual terms are zero because $\mathbb{E}(Z_y - m_{0,y} | W) = 0$. Therefore

$$\text{Cov}\{U_y(a), U_z(a) | W\} = \frac{C_{yz}(W)}{\pi(W)} + (\pi(W)^{-1} - 1)\delta_y^a \delta_z^a. \quad (\text{S15})$$

The conditional means are $m_{0,y}$. Taking total covariance and multiplying by the kernel factors shows that the influence covariance exceeds the efficient one by

$$\mu_h^{-2} \mathbb{E}\{K_h^2(\pi^{-1} - 1)\delta_y^a \delta_z^a\}. \quad (\text{S16})$$

Over every finite grid this is positive semidefinite. If an estimator attains the true m_0 , it dominates other fixed augmentations in covariance and in centred convex Gaussian-set probabilities. The fitted scalar path in the separation example need not contain that estimator; this is why the example does not conflict with operator-efficiency results such

as Zhang and Li (2026).

S8.2 The integrated-variance rule as a comparison method

Put $A = m_0 - q$, $D = m - q$, and use a finite threshold measure ν . The augmentation-dependent integrated variance is

$$J_h(\lambda) = \mathbb{E}\{v_h^*(W)\|A - \lambda D\|_\nu^2\}, \quad v_h^*(W) = \frac{h^d K_h^2}{\mu_h^2}(\pi^{-1} - 1).$$

Direct expansion gives $J_h(\lambda) = J_h(0) - 2\lambda G_h + \lambda^2 H_h$, where

$$G_h = \mathbb{E}\{v_h^*\langle A, D \rangle_\nu\}, \quad H_h = \mathbb{E}\{v_h^*\|D\|_\nu^2\}.$$

Since $\mathbb{E}\{R\pi^{-1}(Z - q) \mid W\} = A$, G_h has an observable inverse-probability representation. For $H_h > 0$ the constrained optimum is $\Pi_{[0,1]}(G_h/H_h)$; for $H_h = 0$ use zero. Full borrowing improves the anchor exactly when $2G_h - H_h > 0$. The improvement is zero when $G_h \leq 0$, G_h^2/H_h when $0 < G_h < H_h$, and $2G_h - H_h$ when $G_h \geq H_h > 0$.

These are retained as efficiency identities and motivate the Trace comparator; they are not counted as new paired-certification principles. Covariance-based trace estimation in the new simulation targets the trace of the full fitted-path covariance, including all terms independent of the weight. Its population minimiser agrees with the equal-weight version of the integrated criterion.

For the earlier ratio selector, bounded moments give normalised estimates $p(\widehat{G}_h - G_h), p(\widehat{H}_h - H_h) = O_P(N_G^{-1/2})$. If pH_h is bounded below, the projection map and a ratio expansion imply $\widehat{\lambda} - \lambda^{\text{or}} = O_P(N_G^{-1/2})$. The correction in an independent evaluation mean has variance of order $(\widehat{\lambda} - \lambda^{\text{or}})^2$ on the local process scale; it is therefore $o_P(1)$. If the correction collapses in the uniform local verification norm, every bounded difference of weights has the same negligible evaluation effect. With an entropy bound this argument extends from finite thresholds to the whole process. It does not require treating a random

fitted regression as the true conditional distribution.

S8.3 Local covariance limit and point-target bias

Suppose $h \rightarrow 0$, $p \rightarrow p_\infty \in [0, 1]$, $e(W)$ converges locally to $e_0(W)$, and the fitted limiting augmentation is a_∞ . Write $\delta = a_\infty - m_0$. Kernel change of variables, uniform local covariance convergence and dominated convergence give the covariance of the $\sqrt{np h^d}$ -scaled ratio process:

$$\begin{aligned} \Gamma(y, z) = & \frac{\kappa_K}{f_X(x_0)} \mathbb{E} \left[C_{yz}/e_0 + (e_0^{-1} - p_\infty) \delta_y \delta_z \right. \\ & \left. + p_\infty \{m_{0,y} - F_0(y | x_0)\} \{m_{0,z} - F_0(z | x_0)\} \mid X = x_0 \right], \end{aligned} \quad (\text{S17})$$

where $\kappa_K = \int K^2 / (\int K)^2$. To see the normalisation, write $\mu_h = h^d f_X(x_0) \int K + o(h^d)$ and $\mathbb{E} K_h^2 b(W) = h^d f_X(x_0) \int K^2 \mathbb{E} \{b(W) \mid X = x_0\} + o(h^d)$. The three terms follow from conditional covariance plus covariance of the conditional mean. Verification scarcity affects the rate and removes the last term when $p_\infty = 0$; selective allocation remains through e_0^{-1} .

For a symmetric second-order kernel, a twice differentiable covariate density and conditional distribution with uniformly bounded second derivatives,

$$\mathbb{E} K_h Z_y = h^d \int K(u) F_0(y | x_0 + hu) f_X(x_0 + hu) du.$$

Taylor expansion makes first-order terms vanish by symmetry. Expanding the denominator in the same way yields $\sup_y |F_h(y) - F_0(y | x_0)| = O(h^2)$. Hence $\sqrt{np h^d} h^2 \rightarrow 0$ transfers the centred process limit and band to the point target. A bias-corrected alternative must also estimate and include the variability of its correction.

If F_0 is strictly increasing and has continuous density bounded away from zero on the quantile range, an ordinary inverse-map expansion gives

$$\sqrt{np h^d} (\hat{Q}(\tau) - Q_0(\tau)) = - \frac{\sqrt{np h^d} \{\hat{F}(Q_0(\tau)) - F_0(Q_0(\tau))\}}{f_{Y|X}(Q_0(\tau) | x_0)} + o_P(1)$$

uniformly in an interior compact probability interval. Rearrangement is first-order neutral under these strict-monotonicity conditions (Chernozhukov et al., 2010). At atoms, use band inversion from S4 rather than this density formula.

S8.4 Estimated verification probabilities

For a separately fitted $\hat{\pi}$ and a fixed candidate a , the exact conditional drift is

$$\mathbb{E} \left[a + \frac{R}{\hat{\pi}}(Z - a) - Z \middle| W \right] = \frac{\hat{\pi} - \pi}{\hat{\pi}}(a - m_0). \quad (\text{S18})$$

For local normalised averaging, put

$$r_{\pi}^2 = \mu_h^{-1} \mathbb{E} \left[K_h \{ (\hat{\pi} - \pi) / \hat{\pi} \}^2 \right], \quad r_a^2 = \mu_h^{-1} \mathbb{E} [K_h(a - m_0)^2].$$

Cauchy–Schwarz bounds the local drift by $r_{\pi}r_a$. A sufficient bias condition for pointwise inference is $\sqrt{np}h^d r_{\pi}r_a = o_P(1)$, uniformly over the reported thresholds. The empirical-process replacement additionally requires the local weighted second moment of the signal difference to vanish, for example

$$\frac{ph^d}{\mu_h^2} \mathbb{E} \left[K_h^2 \pi \{ \hat{\pi}^{-1} - \pi^{-1} \}^2 (Z - a)^2 \right] = o_P(1),$$

with an appropriate entropy condition for uniform inference. Positivity bounds must hold for $\hat{\pi}/\pi$. A learned propensity also affects covariance-slope calibration, requiring its estimated influence or a nuisance remainder negligible on that calibration scale. The primary simulations and EPA analysis use known probabilities. The training-fold logistic sensitivity in Section S10 additionally measures plug-in performance; it does not treat propensity-model correctness alone as proof of the required drift and calibration-score conditions.

S8.5 Local linear estimation and verification allocation

A local linear alternative uses $V_h = (1, (X - x_0)^\top/h)^\top$, $Q_h = \mathbb{E}K_h V_h V_h^\top$ and $\beta_h(y) = Q_h^{-1} \mathbb{E}K_h V_h Z_y$. Its intercept influence for fixed h is

$$e_1^\top Q_h^{-1} K_h V_h \{U_y(a) - V_h^\top \beta_h(y)\}.$$

This follows by differentiating the weighted normal equation; the derivative of Q_h^{-1} supplies the fitted-linear-term subtraction. The same joint path-covariance and paired-gain construction applies when Q_h is nonsingular. Local linear equivalent weights can be negative, so the finite-resolution intercept need not be a cumulative distribution. Monotone whole-distribution envelopes apply to the point target only after the appropriate localisation-bias control establishes grid coverage for that target.

Finally, for an augmentation fixed independently of the audit design, the part of an integrated variance depending on the verification law is of the form $\mathbb{E}[A(W)/\pi(W)]$, $A \geq 0$. Under $\mathbb{E}\pi = p$, Cauchy–Schwarz gives $\mathbb{E}[A/\pi] \geq \{\mathbb{E}\sqrt{A}\}^2/p$, with equality at $\pi \propto \sqrt{A}$ when unconstrained. Under lower and upper probability bounds the pointwise Karush–Kuhn–Tucker equation gives the clipped square-root allocation, with its constant chosen to satisfy the budget. This is an allocation extension of the original variance criterion, not an assertion that it minimises a simultaneous critical value. The present work fixes the known verification design and certifies borrowing conditional on that design.

S9 Reproduction, numerical diagnostics and complete results

S9.1 Frozen protocols and provenance

The source archive is pinned to commit `70c47c5`; the complete identifier is recorded in the source manifest. The date mapping was transferred losslessly: its 43,937 bytes have Git

blob hash `aaab1595ce2f0848783ff4015009ac6218caf5d3`. The reconstructed analysis table preserves all 2,814 supplied observation identifiers, concentrations and humidities. Joining the mapping produces no missing dates or duplicate station–date pairs. The recovery script supplied by the authors checked the three numerical columns and station identity against the archived RDS before extracting its dates. The reproduction package records these source checks separately from the new analyses.

The revised primary simulation contains 6,000 independently generated replications and 36,000 method-level records. Every replication ran successfully; no failed-information fallback occurred in the primary grid. The code keeps the earlier exploratory simulation in `results/prior_checkpoint`; it is not pooled with the guarded simulation. New results reside in `results/guarded_main`. The exact binary calculations and original technical note are retained with their own provenance.

The simulation verification law is random or

$$\pi(x, s) = p \left[1 + 0.4 \tanh \left\{ \frac{s - \ell(x)\mu(x)}{\sqrt{\ell(x)^2\sigma(x)^2 + \tau^2}} \right\} \right].$$

The conditional standardised surrogate has a symmetric distribution, so this design has mean p without using reference outcomes to calibrate the probability. Mean regression uses $1, x, x^2$ for the anchor and adds s for the surrogate candidate; residual standard deviation is estimated within the training role. The anchor fit uses inverse-probability weights and the surrogate fit uses equal weights. Their outputs are bounded Gaussian distribution predictions even under misspecification. The matrix-projection comparator uses the stated fixed relative ridge and is not restricted to the scalar path.

All reported primary bands cover the raw estimator on the declared grid. Integrated squared error is also computed for this raw grid estimator. This differs from earlier archived comparisons of rearranged point estimates. The inferential and risk targets are recorded in the manifest, rather than inferred from a figure label.

S9.2 Numerical accuracy and software checks

The radial Gaussian derivative is checked by central finite differences in dimensions one, two and five. The largest relative discrepancy is below 6×10^{-10} . The two-threshold separation calculation is compared against deterministic Gaussian rectangle integration; the largest discrepancy in the four tested critical values is below 9×10^{-7} . An independent perturbation of the empirical distribution verifies the covariance-centring influence term to absolute error below 7×10^{-11} .

The new numerical audit compares 8,192 and 32,768 radial directions in 60 fitted-data replications. All 60 paired selections agree. The largest change in a radial critical value is 3.935×10^{-4} ; the largest change in a paired radial gain is 4.475×10^{-4} . Refining population-covariance integration from 48 by 24 quadrature points to 96 by 64 changes a covariance entry by at most 2.662×10^{-5} . These are deterministic refinement checks on the evaluated cases. The fresh Gaussian order-statistic intervals, not these observed differences, provide the numerical probability guarantees used by the selector.

Thirty additional revision checks verify date provenance, historical/audit separation, concentration reconstruction, exact binomial quantile ranks, positive-semidefinite covariance handling, the unique-reference budget identity, exact finite-frame covariance enumeration, unbiased gate moments, gate-score variance and whole-distribution envelopes at atoms. Hidden reference outcomes are changed while keeping all visible variables and masks fixed; the EPA selected weights, point estimates and confidence limits remain unchanged. The code treats materially negative eigenvalues as an error and clips only negative values within its stated floating-point tolerance. All checks pass.

An observed zero harmful-selection count is not a probability guarantee. For zero events in 500 independent replications, the exact two-sided 95% binomial upper limit is approximately 0.00735; the one-sided 95% upper limit is approximately 0.00597. Machine-readable exact intervals accompany every simulation coverage and numerical-harm entry. The primary harm threshold is 10^{-4} in the normalised population Gaussian critical value, and the quadrature audit quantifies, but does not rigorously bound, its residual assessment

error.

S9.3 Three-role rotation

A separate 400-replication experiment allocates error across three role-specific bands. Each evaluation role uses Gaussian tail probability $0.047/3$ plus numerical error 0.001, for a total no greater than 0.05 across rotations. Gate certificates allocate $0.047/3$ to statistical calibration and 0.001 to the two numerical events. There are 200 aligned and 200 locally reversed replications.

In the aligned design, the intersection band has coverage 0.960 for Anchor and 0.970 for Paired, with mean widths 0.264 and 0.202. Under local reversal both methods have coverage 0.975 and mean width 0.263. No empty intersections occur. Convex aggregation covers in every observed replication and is wider: aligned widths are 0.358 and 0.273, respectively. These results illustrate the price of dependence-valid rotation without relying on a pooled Gaussian approximation for jointly selected role estimates.

S9.4 EPA decisions and reference support

The six EPA design–budget combinations contain 1,200 audit masks and 18,000 method–target records, with no computation failures. The complete audit frame is held fixed. Monte Carlo coverage variation is therefore variation over designed reference selection, not environmental population sampling.

At 30% humidity and concentration 5, the complete-frame exceedance discrepancy is 18.602 percentage points. Under random auditing, Paired’s simultaneous interval resolves its positive sign in 1.0%, 9.5% and 68.0% of masks at the 5%, 10% and 20% budgets. At concentration 15 the smaller upper-tail differences are generally unresolved: at the 20% random budget, the resolution rates are 0%, 0.5% and 0% at the three humidity targets. This separates a precise description available with all reference measurements from what a sparse audit can establish. It does not convert absence of a resolved sign into evidence of exact calibration.

All pointwise decisions are recovered from the retained selected weights, evaluation widths and reference-mask seeds, without rerunning a selector or choosing new thresholds. Their reconstructed estimates and limits match the stored example bands to floating-point precision. The resulting 126,000 pointwise records and all sign-resolution summaries are retained in the code archive. For an evaluation neighbourhood with fewer than three reference values, the procedure reports the full feasible distribution band translated by the known sensor distribution; all such cases are explicitly flagged.

The next two tables provide every main simulation and EPA comparison. Monte Carlo standard errors for additional metrics and exact binomial intervals are in the machine-readable tables. The full-reference legacy VitalDB and controlled synthetic review results remain in the original project archive; they are not represented as evaluations of the new paired selector.

Table S1: Complete primary simulation comparisons: 500 replications per method and cell.

Cell	Method	Coverage	MCSE	Width ratio	Harm	Adoption
1	Anchor	0.958	0.009	1.000	0.000	0.000
1	Full	0.958	0.009	0.722	0.000	1.000
1	Paired	0.954	0.009	0.803	0.000	0.862
1	Plug-in	0.960	0.009	0.723	0.000	1.000
1	Projection	0.956	0.009	0.750	0.000	—
1	Trace	0.960	0.009	0.721	0.000	1.000
2	Anchor	0.956	0.009	1.000	0.000	0.000
2	Full	0.960	0.009	0.722	0.000	1.000
2	Paired	0.956	0.009	0.747	0.000	1.000
2	Plug-in	0.962	0.009	0.723	0.000	1.000
2	Projection	0.958	0.009	0.734	0.000	—
2	Trace	0.962	0.009	0.722	0.000	1.000

Cell	Method	Coverage	MCSE	Width ratio	Harm	Adoption
3	Anchor	0.962	0.009	1.000	0.000	0.000
3	Full	0.956	0.009	0.759	0.000	1.000
3	Paired	0.958	0.009	0.769	0.000	1.000
3	Plug-in	0.954	0.009	0.759	0.000	1.000
3	Projection	0.966	0.008	0.764	0.000	—
3	Trace	0.954	0.009	0.759	0.000	1.000
4	Anchor	0.952	0.010	1.000	0.000	0.000
4	Full	0.966	0.008	0.723	0.000	1.000
4	Paired	0.956	0.009	0.831	0.000	0.752
4	Plug-in	0.968	0.008	0.724	0.000	1.000
4	Projection	0.980	0.006	0.761	0.000	—
4	Trace	0.970	0.008	0.722	0.000	1.000
5	Anchor	0.936	0.011	1.000	0.000	0.000
5	Full	0.958	0.009	0.715	0.000	1.000
5	Paired	0.952	0.010	0.750	0.000	0.994
5	Plug-in	0.956	0.009	0.715	0.000	1.000
5	Projection	0.966	0.008	0.733	0.000	—
5	Trace	0.956	0.009	0.714	0.000	1.000
6	Anchor	0.970	0.008	1.000	0.000	0.000
6	Full	0.958	0.009	0.738	0.000	1.000
6	Paired	0.962	0.009	0.755	0.000	1.000
6	Plug-in	0.958	0.009	0.739	0.000	1.000
6	Projection	0.958	0.009	0.746	0.000	—
6	Trace	0.960	0.009	0.739	0.000	1.000
7	Anchor	0.960	0.009	1.000	0.000	0.000
7	Full	0.966	0.008	0.954	0.000	1.000

Cell	Method	Coverage	MCSE	Width ratio	Harm	Adoption
7	Paired	0.960	0.009	1.000	0.000	0.000
7	Plug-in	0.962	0.009	0.955	0.000	0.998
7	Projection	0.960	0.009	0.966	0.014	—
7	Trace	0.964	0.008	0.955	0.000	1.000
8	Anchor	0.954	0.009	1.000	0.000	0.000
8	Full	0.958	0.009	1.114	0.994	1.000
8	Paired	0.954	0.009	1.000	0.000	0.000
8	Plug-in	0.954	0.009	1.000	0.000	0.006
8	Projection	0.966	0.008	0.805	0.000	—
8	Trace	0.954	0.009	1.000	0.000	0.006
9	Anchor	0.968	0.008	1.000	0.000	0.000
9	Full	0.968	0.008	1.001	0.694	1.000
9	Paired	0.968	0.008	1.000	0.000	0.000
9	Plug-in	0.966	0.008	1.000	0.312	0.476
9	Projection	0.958	0.009	1.013	1.000	—
9	Trace	0.966	0.008	1.000	0.310	0.492
10	Anchor	0.944	0.010	1.000	0.000	0.000
10	Full	0.944	0.010	1.000	0.000	1.000
10	Paired	0.944	0.010	1.000	0.000	0.000
10	Plug-in	0.944	0.010	1.000	0.000	0.000
10	Projection	0.944	0.010	1.000	0.000	—
10	Trace	0.944	0.010	1.000	0.000	0.000
11	Anchor	0.958	0.009	1.000	0.000	0.000
11	Full	0.964	0.008	0.721	0.000	1.000
11	Paired	0.968	0.008	0.731	0.000	1.000
11	Plug-in	0.966	0.008	0.721	0.000	1.000

Cell	Method	Coverage	MCSE	Width ratio	Harm	Adoption
11	Projection	0.966	0.008	0.727	0.000	–
11	Trace	0.966	0.008	0.721	0.000	1.000
12	Anchor	0.956	0.009	1.000	0.000	0.000
12	Full	0.954	0.009	1.135	1.000	1.000
12	Paired	0.956	0.009	1.000	0.000	0.000
12	Plug-in	0.956	0.009	1.000	0.002	0.002
12	Projection	0.954	0.009	0.839	0.004	–
12	Trace	0.956	0.009	1.000	0.002	0.002

Table S2: Targetwise EPA design-based performance. R and S denote random and selective audit; the following number is the budget percentage.

Design	RH	Method	Coverage	Width ratio	Weight	Harm
R5	30	Anchor	0.985	1.000	0.000	0.000
R5	30	Full	0.995	0.653	1.000	0.000
R5	30	Paired	0.985	0.996	0.011	0.000
R5	30	Plug-in	0.995	0.676	0.897	0.000
R5	30	Trace	0.995	0.663	0.947	0.000
R5	50	Anchor	0.980	1.000	0.000	0.000
R5	50	Full	0.995	0.690	1.000	0.000
R5	50	Paired	0.980	0.975	0.072	0.000
R5	50	Plug-in	0.995	0.699	0.882	0.000
R5	50	Trace	0.995	0.695	0.919	0.000
R5	70	Anchor	0.990	1.000	0.000	0.000
R5	70	Full	1.000	0.694	1.000	0.000
R5	70	Paired	0.990	0.985	0.048	0.000
R5	70	Plug-in	1.000	0.712	0.865	0.000

Design	RH	Method	Coverage	Width ratio	Weight	Harm
R5	70	Trace	1.000	0.702	0.912	0.000
R10	30	Anchor	0.985	1.000	0.000	0.000
R10	30	Full	0.990	0.646	1.000	0.000
R10	30	Paired	0.990	0.980	0.051	0.000
R10	30	Plug-in	0.990	0.657	0.927	0.000
R10	30	Trace	0.990	0.649	0.972	0.000
R10	50	Anchor	0.980	1.000	0.000	0.000
R10	50	Full	0.990	0.678	1.000	0.000
R10	50	Paired	0.980	0.899	0.226	0.000
R10	50	Plug-in	0.990	0.682	0.902	0.000
R10	50	Trace	0.990	0.679	0.934	0.000
R10	70	Anchor	0.990	1.000	0.000	0.000
R10	70	Full	0.990	0.677	1.000	0.000
R10	70	Paired	0.990	0.948	0.134	0.000
R10	70	Plug-in	0.990	0.684	0.906	0.000
R10	70	Trace	0.985	0.679	0.933	0.000
R20	30	Anchor	0.990	1.000	0.000	0.000
R20	30	Full	0.995	0.637	1.000	0.000
R20	30	Paired	0.985	0.872	0.302	0.000
R20	30	Plug-in	0.990	0.644	0.955	0.000
R20	30	Trace	0.995	0.638	0.987	0.000
R20	50	Anchor	0.990	1.000	0.000	0.000
R20	50	Full	1.000	0.670	1.000	0.000
R20	50	Paired	0.995	0.758	0.559	0.000
R20	50	Plug-in	1.000	0.672	0.926	0.000
R20	50	Trace	1.000	0.670	0.959	0.000

Design	RH	Method	Coverage	Width ratio	Weight	Harm
R20	70	Anchor	0.990	1.000	0.000	0.000
R20	70	Full	0.980	0.671	1.000	0.000
R20	70	Paired	0.990	0.855	0.349	0.000
R20	70	Plug-in	0.980	0.673	0.943	0.000
R20	70	Trace	0.980	0.672	0.951	0.000
S5	30	Anchor	0.965	1.000	0.000	0.000
S5	30	Full	1.000	0.676	1.000	0.000
S5	30	Paired	0.965	1.000	0.000	0.000
S5	30	Plug-in	0.990	0.696	0.876	0.000
S5	30	Trace	0.985	0.683	0.925	0.000
S5	50	Anchor	0.990	1.000	0.000	0.000
S5	50	Full	0.995	0.671	1.000	0.000
S5	50	Paired	0.990	0.967	0.077	0.000
S5	50	Plug-in	0.995	0.682	0.895	0.000
S5	50	Trace	0.995	0.678	0.927	0.000
S5	70	Anchor	1.000	1.000	0.000	0.000
S5	70	Full	0.995	0.647	1.000	0.000
S5	70	Paired	1.000	0.998	0.013	0.000
S5	70	Plug-in	0.995	0.669	0.864	0.000
S5	70	Trace	0.995	0.656	0.921	0.000
S10	30	Anchor	0.960	1.000	0.000	0.000
S10	30	Full	0.990	0.649	1.000	0.000
S10	30	Paired	0.960	0.994	0.011	0.000
S10	30	Plug-in	0.990	0.668	0.915	0.000
S10	30	Trace	0.990	0.654	0.960	0.000
S10	50	Anchor	0.980	1.000	0.000	0.000

Design	RH	Method	Coverage	Width ratio	Weight	Harm
S10	50	Full	0.995	0.651	1.000	0.000
S10	50	Paired	0.980	0.890	0.234	0.000
S10	50	Plug-in	0.995	0.657	0.908	0.000
S10	50	Trace	0.995	0.655	0.937	0.000
S10	70	Anchor	0.985	1.000	0.000	0.000
S10	70	Full	0.995	0.656	1.000	0.000
S10	70	Paired	0.985	0.966	0.070	0.000
S10	70	Plug-in	0.995	0.661	0.901	0.000
S10	70	Trace	0.995	0.656	0.932	0.000
S20	30	Anchor	0.975	1.000	0.000	0.000
S20	30	Full	0.990	0.647	1.000	0.000
S20	30	Paired	0.975	0.935	0.166	0.000
S20	30	Plug-in	0.985	0.663	0.930	0.000
S20	30	Trace	0.985	0.652	0.970	0.000
S20	50	Anchor	0.985	1.000	0.000	0.000
S20	50	Full	1.000	0.646	1.000	0.000
S20	50	Paired	0.985	0.746	0.558	0.000
S20	50	Plug-in	1.000	0.649	0.939	0.000
S20	50	Trace	1.000	0.647	0.962	0.000
S20	70	Anchor	0.995	1.000	0.000	0.000
S20	70	Full	0.995	0.652	1.000	0.000
S20	70	Paired	0.995	0.888	0.240	0.000
S20	70	Plug-in	0.990	0.654	0.923	0.000
S20	70	Trace	0.990	0.652	0.947	0.000

S10 Dense thresholds, matched certification and precision requirements

S10.1 Experiments fixed before the additional runs

The additional protocol contains three modules: six pairing-ablation cells with 500 replications each, six dense-grid cells with 200 replications each, and seven robustness cells with 300 replications each. This gives 6,300 design-grid replications and 37,800 method records. The dense-grid cells deliberately reuse each generated dataset at three grid resolutions: they contain 400 distinct datasets, not 1,200 independent datasets. The additional modules together use 5,500 distinct generated datasets. All six methods within a replication share training, reference masks, gate and evaluation data. These runs are separate from the 6,000 primary replications.

The ablation repeats alignment at reference fractions 0.05, 0.10, 0.20 and weak, reversed and absent signals at fraction 0.10, always with $n = 6,000$, $h = 0.35$ and seven thresholds. Unpaired uses the absolute-critical-value bounds of Section S3.5, with statistical errors 0.0245 for the anchor and for the candidate maximum. The two upper simulated quantiles each receive error 0.00025. Its Gaussian numerical intervals are exactly the same intervals used by Paired and have total error 0.0005. Thus each selector has total statistical-plus-numerical error 0.05. Each absolute Taylor expansion receives the same b_N cushion; the sum is $2b_N$, compared with Paired's λb_N . This is part of the error-cancellation comparison, not a different reference budget. The other four methods are Anchor, Full, Trace and Plug-in. The matrix Projection comparator remains in the primary experiment.

The dense experiments use $J = 21, 51, 101$ thresholds $0.8\Phi^{-1}(u_j)$ with evenly spaced $u_j \in [0.02, 0.98]$. The 21- and 51-point grids are subsets of the 101-point grid. Both alignment with random verification and local reversal with selective verification use $n = 6,000$, $p = 0.10$ and $h = 0.35$. A common enlargement $10^{-8}I_J$ is declared for these calculations. Every reference mask, candidate regression and generated dataset is

shared across grid resolutions. We evaluate the raw simultaneous band on its grid and its monotone envelope on a common 1,001-point grid. The latter is a numerical check of the envelope; the all-threshold guarantee follows from monotonicity as proved in Section S4. The experiment does not replace a growing-grid regularity condition by an empirical coverage estimate.

S10.2 Outcome and nuisance-regression robustness

The three continuous non-Gaussian designs replace the normal error by a standardised t_3 , $(\chi_3^2 - 3)/\sqrt{6}$, or $(A + 0.5Z)/\sqrt{1.25}$ with independent $A \in \{-1, 1\}$ equally likely and $Z \sim N(0, 1)$. Mean and scale functions are unchanged. These errors have mean zero and variance one. Candidate Gaussian distribution regressions can be misspecified, while their threshold predictions remain bounded. Surrogate noise is Gaussian with standard deviation 0.4.

For the ordinal design, generate the normal latent outcome and discretise it using nine cutpoints equally spaced from -2 to 2 , producing categories $0, \dots, 9$. Set $S = Y + 0.4\eta$ and evaluate all nine nontrivial cumulative probabilities. True probabilities use the latent normal distribution at the category cutpoints. The two-dimensional design takes independent $X_1, X_2 \sim U[-1, 1]$, mean $1.2X_1 + 0.6X_1^2 + 0.8X_2$, scale $0.8 + 0.25(X_1^2 + X_2^2)$, a product Epanechnikov kernel with $h = 0.5$, and target $(0, 0)$. Its local information is $n_G p h^2$.

A separate nonparametric-candidate design uses empirical neighbour distribution functions: the anchor searches in X , and the surrogate candidate in (X, S) . Coordinates are divided by their verified-training standard deviations, floored at 0.1. The neighbour count is the smaller of the verified-training count and $\max\{20, \lfloor \sqrt{n_{T,R}} \rfloor\}$. All nine nontrivial category thresholds are used for the ordinal design; other robustness cells have 21 thresholds on the same probability range as the dense-grid study. All have $n = 6,000$, $p = 0.10$, and common enlargement $10^{-8}I_J$.

The learned-verification sensitivity uses $\pi(X) = \text{expit}(a + 0.8X)$, with a determined by

$\mathbb{E}\pi(X) = 0.10$. A logistic regression of R on $(1, X)$ is fitted only in the training role; its predictions are clipped to $[0.005, 0.5]$. These fitted probabilities replace the known design probabilities in the downstream signals. This cell measures actual plug-in sensitivity and is reported separately from the known-probability calibration result. Learned-probability inference requires the local drift and score-replacement conditions in Section S8; fitting a correct propensity model alone does not establish those product-rate conditions for every misspecified outcome regression.

For each design, the reference CDF is computed by integrating its known conditional distribution over the kernel neighbourhood, using 100-node Gauss–Legendre quadrature per covariate. Independent refinement to 180 nodes is checked. Population critical-value gains and harmful-selection probabilities are reported only in the normal, one-dimensional, Gaussian-candidate known-design cases where the conditional Gaussian covariance calculation applies. They are left unavailable in the other cells, rather than replacing an exact population comparison by an unlabelled finite truth sample. Coverage, width, distribution loss and adoption remain observed in every cell.

S10.3 Numerical refinement at dense thresholds

The first five replication identifiers in each dense-grid cell are re-evaluated with 8,192 and 32,768 radial directions. These 30 comparisons were fixed before the refinement. Both calculations retain the same numerical critical-value intervals and total confidence budgets. We report changes in critical values, score radii, chosen weights, evaluation widths and point estimates. Weight agreement alone is not a measure of Gaussian integration accuracy when several grid choices have similar lower gains. The delivered records retain every disagreement and its inferential consequences. Independent directional finite differences also check the gradient in dimensions 21, 51 and 101. These calculations validate the implementation of the numerical derivative and measure integration sensitivity; the separate direct-Gaussian order-statistic construction supplies the numerical critical-value probability guarantee.

S10.4 Additional results and interpretation

The completed run contains 6,300 design-grid replications, with no execution failures. Table S3 gives coverage, width and adoption for every additional cell and method. Exact binomial intervals and Monte Carlo standard errors for all continuous summaries accompany the machine-readable table. These are cellwise Monte Carlo intervals, not a simultaneous assertion across all simulation cells. The dense output eigenvalues and local reference counts are retained alongside each replication. The 1,001-point envelope check is separate from grid coverage.

In the paired ablation, width differences and their standard errors are computed within the same generated sample. Two absolute critical-value bounds can remain too wide to certify a gain even when the paired difference can do so. The lower-bound constants and the allocation of error are fully disclosed in Section S3.5; this experiment isolates the stated implementations rather than every possible unpaired optimisation.

In the 30 dense refinement cases, 25 weights agree. Maximum absolute changes in radial critical values and paired radial gains are 0.005813 and 0.004265. Maximum relative changes in an evaluated band width and in the corresponding population critical value are 3.258% and 2.140%. No refined selected candidate in these cases has a negative assessed gain. Selection labels, integration error and inferential impact are therefore all recorded, rather than requiring equality of a mesh index as a proxy for numerical accuracy.

The seven robustness cells retain Gaussian-candidate misspecification or use the declared neighbour estimator. Their true CDFs are evaluated analytically in the outcome dimension and by covariate quadrature; the independent 180-node refinement and implementation tests are archived. These checks do not convert an asymptotic statement into a finite-sample guarantee for an estimated verification model.

S10.5 Extended EPA precision experiment

The original 1,200 budget–design–mask combinations and 126,000 threshold records remain unchanged. The added random-audit grid has 1,600 budget–mask combinations, 24,000 method–humidity records and 168,000 thresholdwise intervals. It uses the unchanged historical candidates and the same mask generator as the original study. The first 200 uniform-draw families are nested with those already reported; the next 200 are additional families. No thresholds are selected by their sign-resolution rates. Table S4 reports all methods on the added budgets. Complete targetwise coverage, false-sign family probabilities, adoption and interval-width summaries are in `results/v3/reporting`.

The first-tested-budget summary distinguishes an observed resolution rate above 0.80 from an exact 95% Monte Carlo lower bound above 0.80. It does not interpolate a minimum required budget. For $p = 1$ the entire audit frame is observed and the contrast is known; this exact census endpoint is not passed through a Gaussian covariance enlargement or represented as an additional simulation. Historical calibration measurements remain an additional cost at every budget.

Table S3: Complete additional validation. Coverage and adoption are frequencies; width is the mean within-replication ratio to Anchor.

Cell	Method	Coverage	MCSE	Width	Adoption
101: aligned, $p = 0.05$					
101	Anchor	0.954	0.009	1.000	0.000
101	Full	0.966	0.008	0.723	1.000
101	Trace	0.968	0.008	0.722	1.000
101	Plug-in	0.968	0.008	0.724	1.000
101	Paired	0.952	0.010	0.802	0.858
101	Unpaired	0.954	0.009	1.000	0.000
102: aligned, $p = 0.10$					
102	Anchor	0.948	0.010	1.000	0.000

Cell	Method	Coverage	MCSE	Width	Adoption
102	Full	0.952	0.010	0.724	1.000
102	Trace	0.954	0.009	0.724	1.000
102	Plug-in	0.958	0.009	0.724	1.000
102	Paired	0.954	0.009	0.747	0.998
102	Unpaired	0.948	0.010	1.000	0.000
103: aligned, $p = 0.20$					
103	Anchor	0.958	0.009	1.000	0.000
103	Full	0.954	0.009	0.759	1.000
103	Trace	0.956	0.009	0.759	1.000
103	Plug-in	0.954	0.009	0.759	1.000
103	Paired	0.956	0.009	0.769	1.000
103	Unpaired	0.958	0.009	0.993	0.032
104: weak, $p = 0.10$					
104	Anchor	0.928	0.012	1.000	0.000
104	Full	0.938	0.011	0.956	1.000
104	Trace	0.938	0.011	0.957	0.998
104	Plug-in	0.934	0.011	0.957	0.998
104	Paired	0.928	0.012	1.000	0.002
104	Unpaired	0.928	0.012	1.000	0.000
105: reversal, $p = 0.10$					
105	Anchor	0.942	0.010	1.000	0.000
105	Full	0.946	0.010	1.110	1.000
105	Trace	0.942	0.010	1.000	0.004
105	Plug-in	0.942	0.010	1.000	0.004
105	Paired	0.942	0.010	1.000	0.000
105	Unpaired	0.942	0.010	1.000	0.000

Cell	Method	Coverage	MCSE	Width	Adoption
106: no signal, $p = 0.10$					
106	Anchor	0.958	0.009	1.000	0.000
106	Full	0.958	0.009	1.002	1.000
106	Trace	0.958	0.009	1.000	0.496
106	Plug-in	0.958	0.009	1.000	0.484
106	Paired	0.958	0.009	1.000	0.000
106	Unpaired	0.958	0.009	1.000	0.000
206: aligned, $J = 21$					
206	Anchor	0.960	0.014	1.000	0.000
206	Full	0.940	0.017	0.740	1.000
206	Trace	0.940	0.017	0.740	1.000
206	Plug-in	0.940	0.017	0.741	1.000
206	Paired	0.955	0.015	0.765	1.000
206	Unpaired	0.960	0.014	1.000	0.000
207: aligned, $J = 51$					
207	Anchor	0.960	0.014	1.000	0.000
207	Full	0.965	0.013	0.750	1.000
207	Trace	0.960	0.014	0.750	1.000
207	Plug-in	0.965	0.013	0.751	1.000
207	Paired	0.965	0.013	0.782	0.985
207	Unpaired	0.960	0.014	1.000	0.000
208: aligned, $J = 101$					
208	Anchor	0.960	0.014	1.000	0.000
208	Full	0.960	0.014	0.753	1.000
208	Trace	0.960	0.014	0.753	1.000
208	Plug-in	0.960	0.014	0.754	1.000

Cell	Method	Coverage	MCSE	Width	Adoption
208	Paired	0.955	0.015	0.798	0.965
208	Unpaired	0.960	0.014	1.000	0.000
209: reversal, $J = 21$					
209	Anchor	0.935	0.017	1.000	0.000
209	Full	0.925	0.019	1.128	1.000
209	Trace	0.935	0.017	1.000	0.010
209	Plug-in	0.935	0.017	1.000	0.005
209	Paired	0.935	0.017	1.000	0.000
209	Unpaired	0.935	0.017	1.000	0.000
210: reversal, $J = 51$					
210	Anchor	0.945	0.016	1.000	0.000
210	Full	0.940	0.017	1.124	1.000
210	Trace	0.945	0.016	1.000	0.010
210	Plug-in	0.945	0.016	1.000	0.010
210	Paired	0.945	0.016	1.000	0.000
210	Unpaired	0.945	0.016	1.000	0.000
211: reversal, $J = 101$					
211	Anchor	0.950	0.015	1.000	0.000
211	Full	0.930	0.018	1.123	1.000
211	Trace	0.950	0.015	1.000	0.010
211	Plug-in	0.950	0.015	1.000	0.010
211	Paired	0.950	0.015	1.000	0.000
211	Unpaired	0.950	0.015	1.000	0.000
301: t3, $d = 1$, gaussian, random					
301	Anchor	0.967	0.010	1.000	0.000
301	Full	0.960	0.011	0.809	1.000

Cell	Method	Coverage	MCSE	Width	Adoption
301	Trace	0.953	0.012	0.807	1.000
301	Plug-in	0.960	0.011	0.807	1.000
301	Paired	0.963	0.011	0.841	0.943
301	Unpaired	0.967	0.010	1.000	0.000
302: skew, $d = 1$, gaussian, random					
302	Anchor	0.960	0.011	1.000	0.000
302	Full	0.960	0.011	0.779	1.000
302	Trace	0.960	0.011	0.778	1.000
302	Plug-in	0.960	0.011	0.779	1.000
302	Paired	0.960	0.011	0.807	0.980
302	Unpaired	0.960	0.011	1.000	0.000
303: mixture, $d = 1$, gaussian, random					
303	Anchor	0.953	0.012	1.000	0.000
303	Full	0.977	0.009	0.723	1.000
303	Trace	0.977	0.009	0.723	1.000
303	Plug-in	0.977	0.009	0.723	1.000
303	Paired	0.970	0.010	0.744	1.000
303	Unpaired	0.953	0.012	1.000	0.000
304: discrete, $d = 1$, gaussian, random					
304	Anchor	0.947	0.013	1.000	0.000
304	Full	0.963	0.011	0.667	1.000
304	Trace	0.960	0.011	0.665	1.000
304	Plug-in	0.960	0.011	0.666	1.000
304	Paired	0.953	0.012	0.686	1.000
304	Unpaired	0.950	0.013	0.994	0.020
305: normal, $d = 2$, gaussian, random					

Cell	Method	Coverage	MCSE	Width	Adoption
305	Anchor	0.937	0.014	1.000	0.000
305	Full	0.957	0.012	0.754	1.000
305	Trace	0.963	0.011	0.754	1.000
305	Plug-in	0.967	0.010	0.755	1.000
305	Paired	0.950	0.013	0.822	0.877
305	Unpaired	0.937	0.014	1.000	0.000
306: normal, $d = 1$, knn, random					
306	Anchor	0.983	0.007	1.000	0.000
306	Full	0.960	0.011	0.753	1.000
306	Trace	0.960	0.011	0.753	1.000
306	Plug-in	0.960	0.011	0.753	1.000
306	Paired	0.963	0.011	0.777	0.993
306	Unpaired	0.983	0.007	1.000	0.000
307: normal, $d = 1$, gaussian, estimated					
307	Anchor	0.960	0.011	1.000	0.000
307	Full	0.970	0.010	0.737	1.000
307	Trace	0.970	0.010	0.737	1.000
307	Plug-in	0.970	0.010	0.738	1.000
307	Paired	0.963	0.011	0.768	0.997
307	Unpaired	0.960	0.011	1.000	0.000

Table S4: Added random EPA budgets. Coverage is joint over 21 threshold–humidity combinations; width is averaged across humidity within each mask and divided by Anchor. Each row uses 400 masks.

p (%)	Method	Joint coverage	MCSE	Width ratio
40	Anchor	0.968	0.009	1.000

p (%)	Method	Joint coverage	MCSE	Width ratio
40	Full	0.973	0.008	0.654
40	Trace	0.975	0.008	0.654
40	Plug-in	0.970	0.009	0.655
40	Paired	0.975	0.008	0.701
60	Anchor	0.963	0.009	1.000
60	Full	0.978	0.007	0.651
60	Trace	0.978	0.007	0.651
60	Plug-in	0.978	0.007	0.652
60	Paired	0.968	0.009	0.671
80	Anchor	0.960	0.010	1.000
80	Full	0.960	0.010	0.650
80	Trace	0.960	0.010	0.650
80	Plug-in	0.960	0.010	0.651
80	Paired	0.960	0.010	0.662
95	Anchor	0.968	0.009	1.000
95	Full	0.945	0.011	0.650
95	Trace	0.943	0.012	0.650
95	Plug-in	0.948	0.011	0.650
95	Paired	0.948	0.011	0.658

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